Cross-Wearer Gesture Recognition — LOUO Variant Comparison

Generated 2026-05-13 23:51:39

Data & configuration

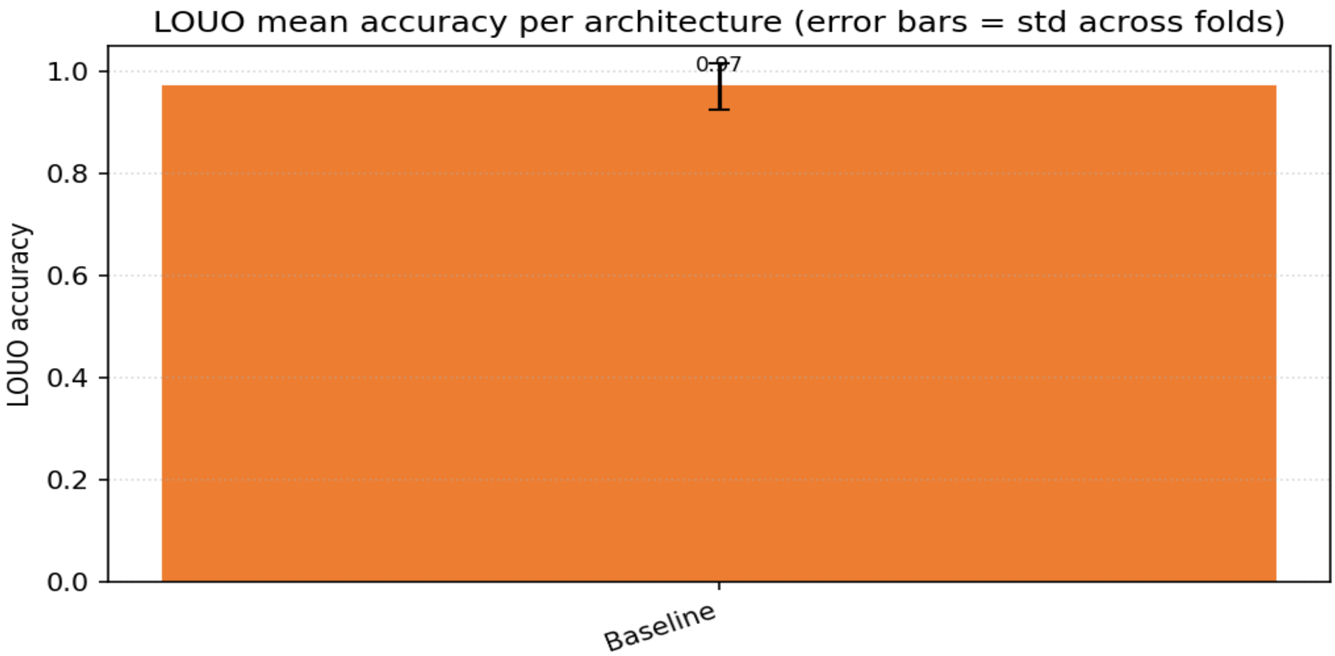
LOUO path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
LOUO path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
LOUO path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
LOUO path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
LOUO path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
LOUO path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
LOUO path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
LOUO path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
LOUO path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
LOUO path 10	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
LOUO path 11	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Dynamic
LOUO path 12	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
LOUO path 13	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
LOUO users loaded	13 (11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry, 8_Jestin)
Excluded classes	Double_Nothing
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 30)
Resample / filter	30 steps · Butterworth cutoff 10.0 Hz, order 4
Per-trial normalisation	zscore
Downstream normalisation	minmax
Sampling rate	30.0 Hz
Random state	42
Augmentation	off
Epochs · batch size	40 · 16
Early stopping	on · monitor val_loss · patience=8
LOUO inner-validation	user
LOUO variants	Baseline
Final-retrain variant	BN_GAP_Wide
Final-retrain held-out user	— (trained on all users)

Variant comparison (LOUO)

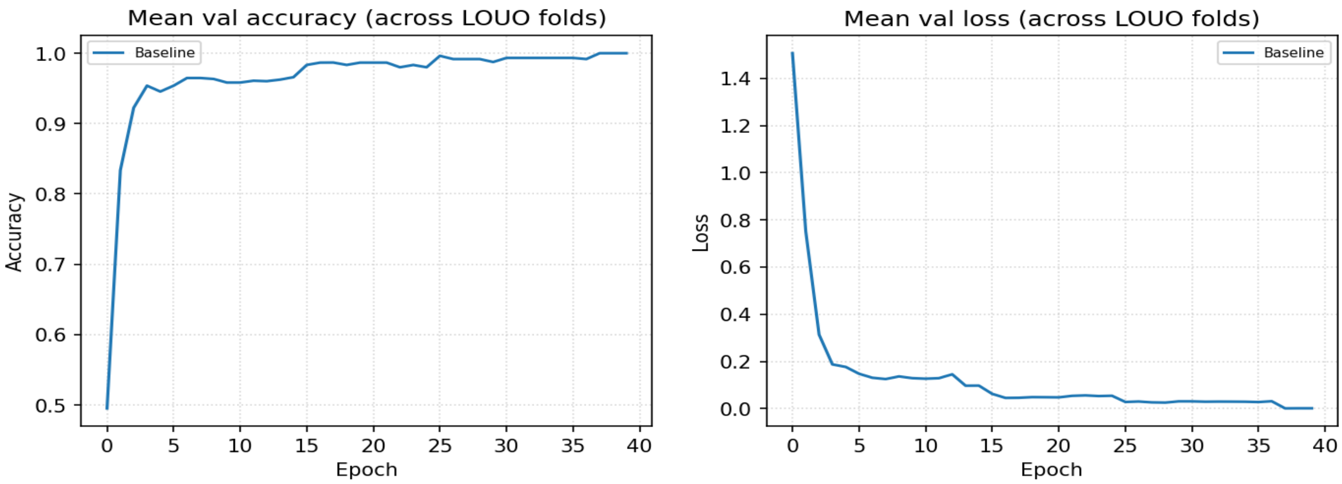
Variant	Params	LOUO mean	LOUO std	Folds	Held-out user acc
Baseline	51,750	0.9718	0.0450	13	—

Best LOUO mean: Baseline

LOUO accuracy per variant

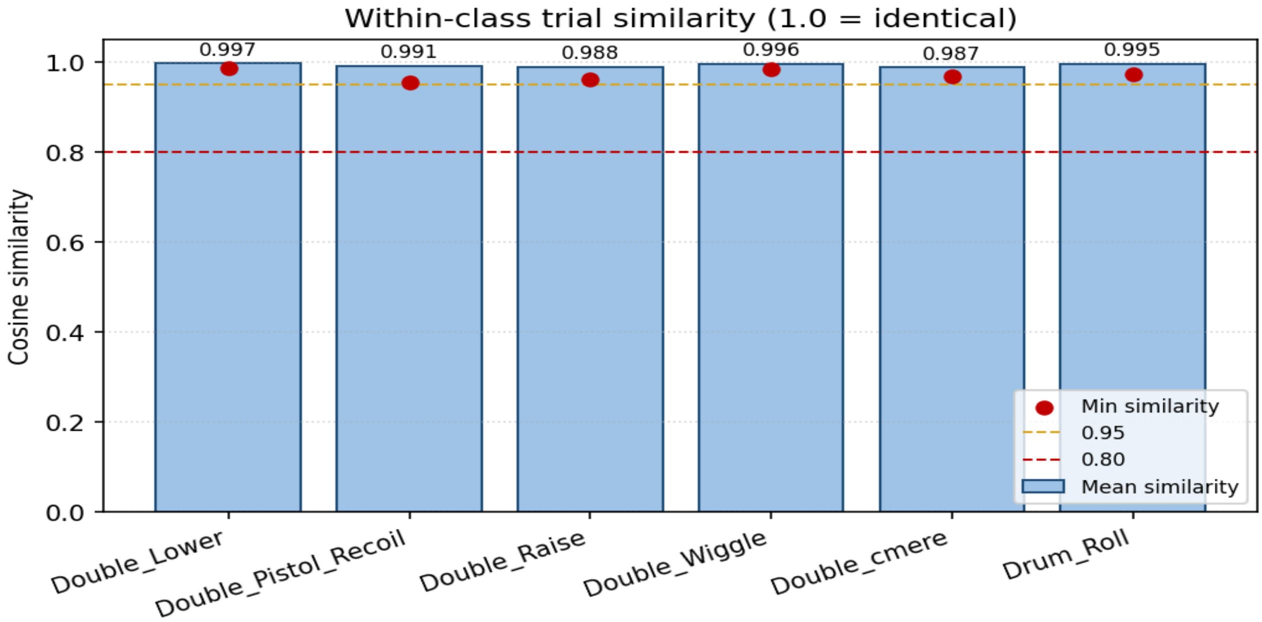


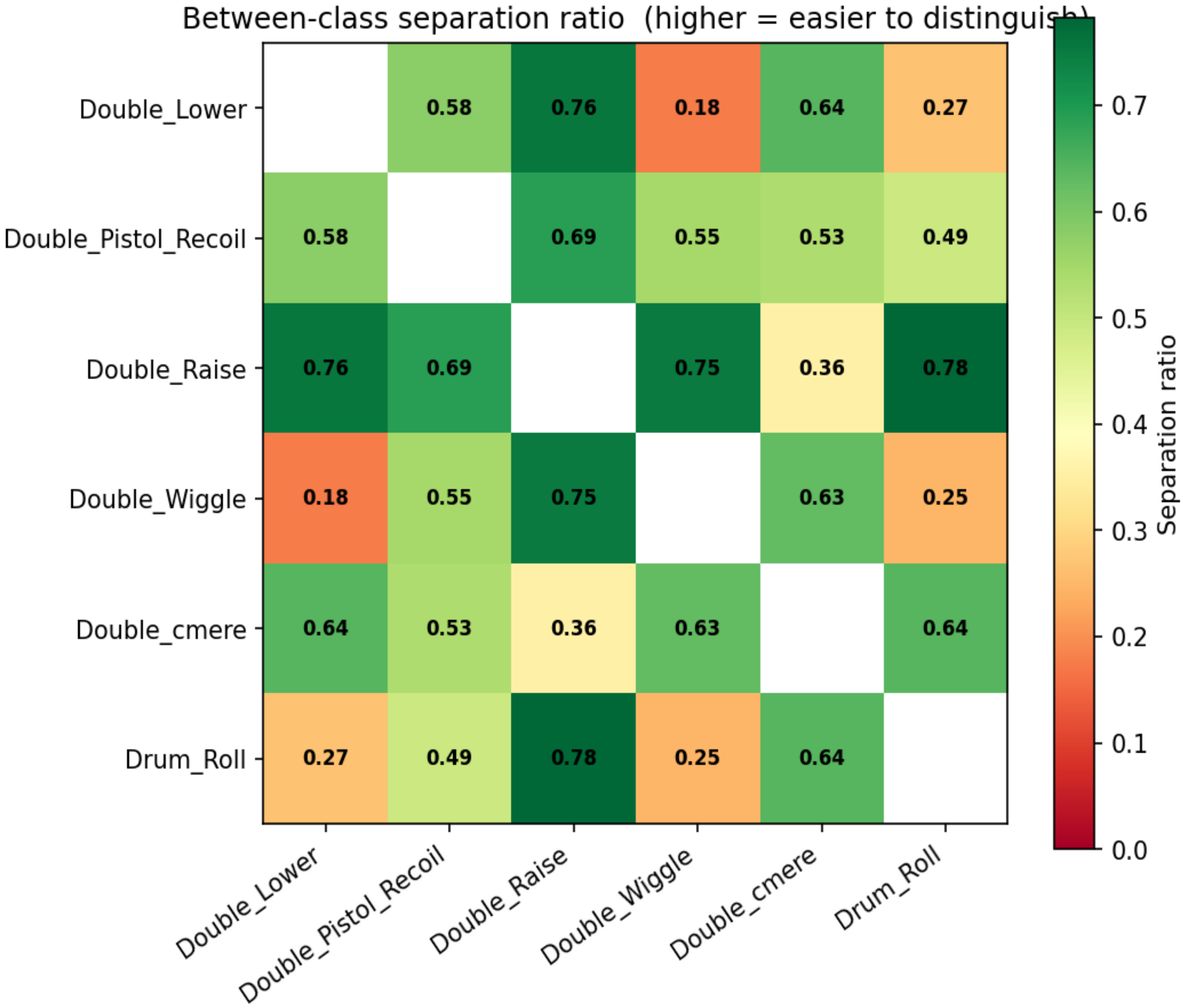
Mean training curves across LOUO folds

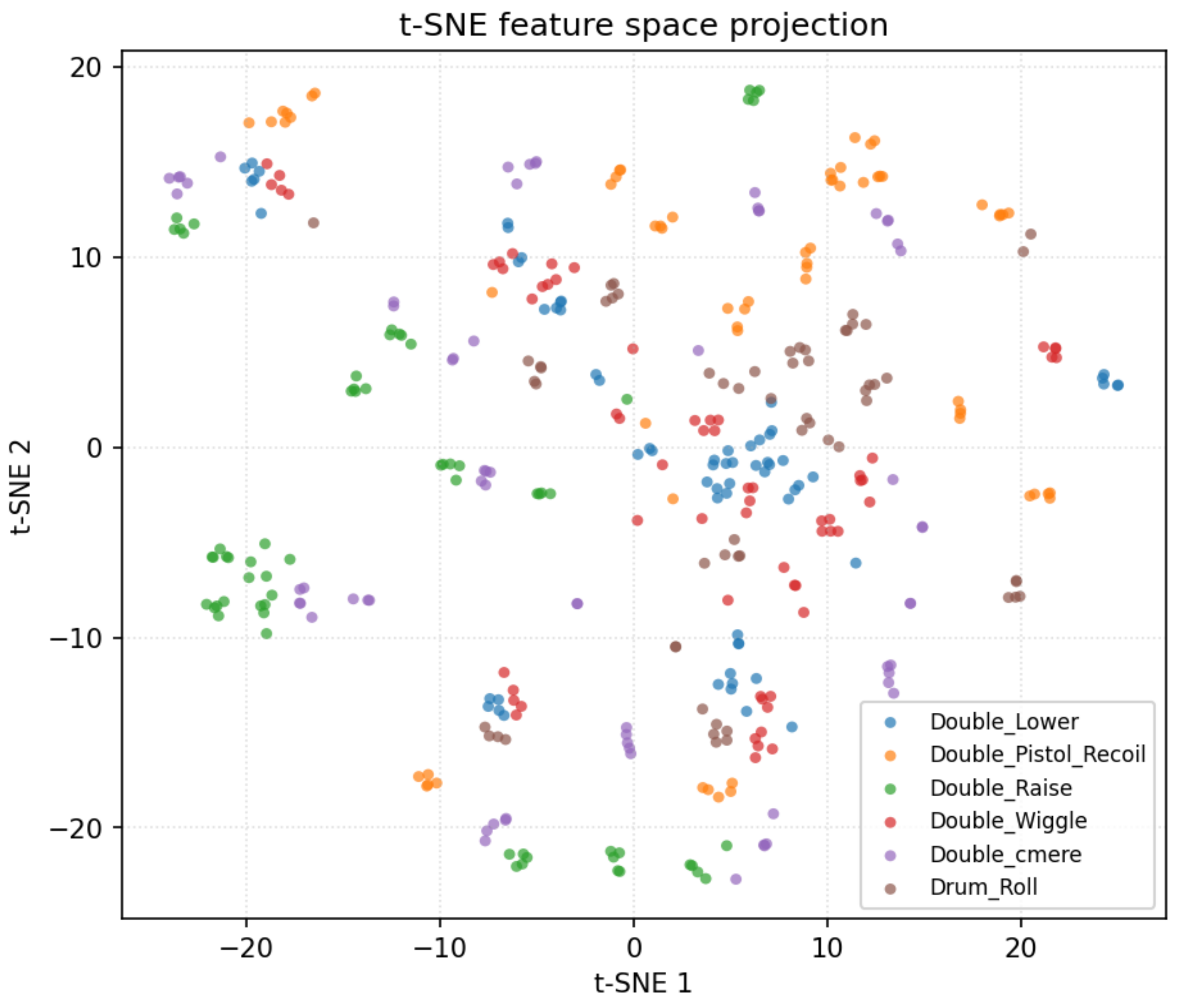


Data diagnostics — pooled across users

Filtered + resampled trials from every LOUO user (before per-trial normalisation).



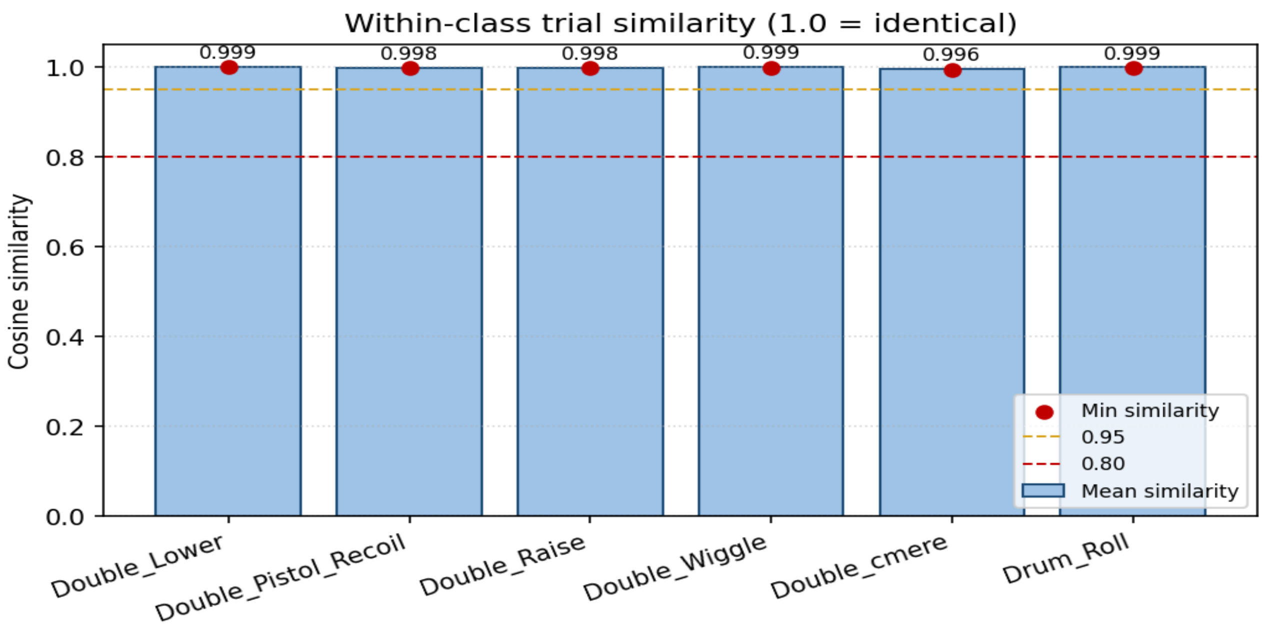


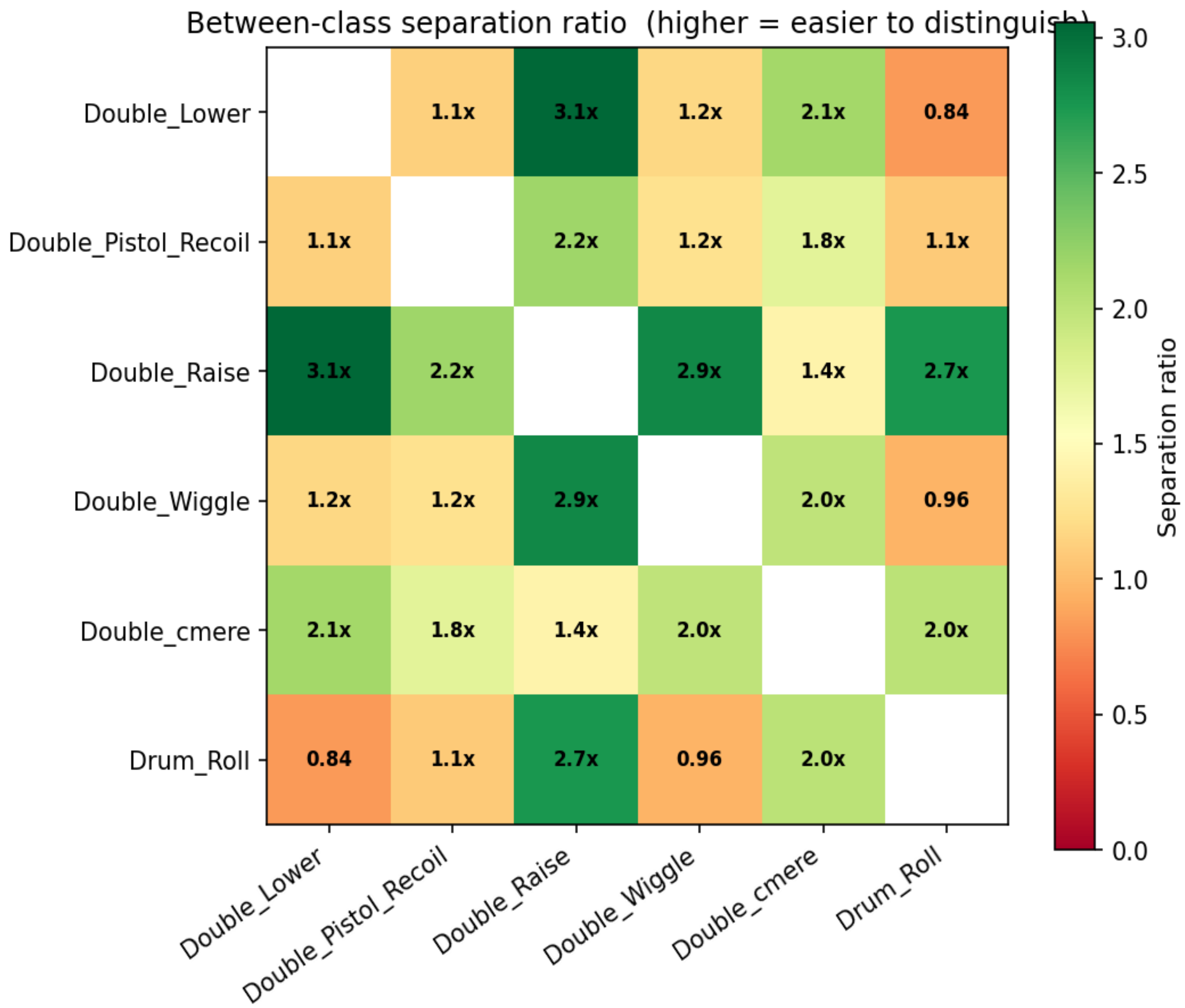


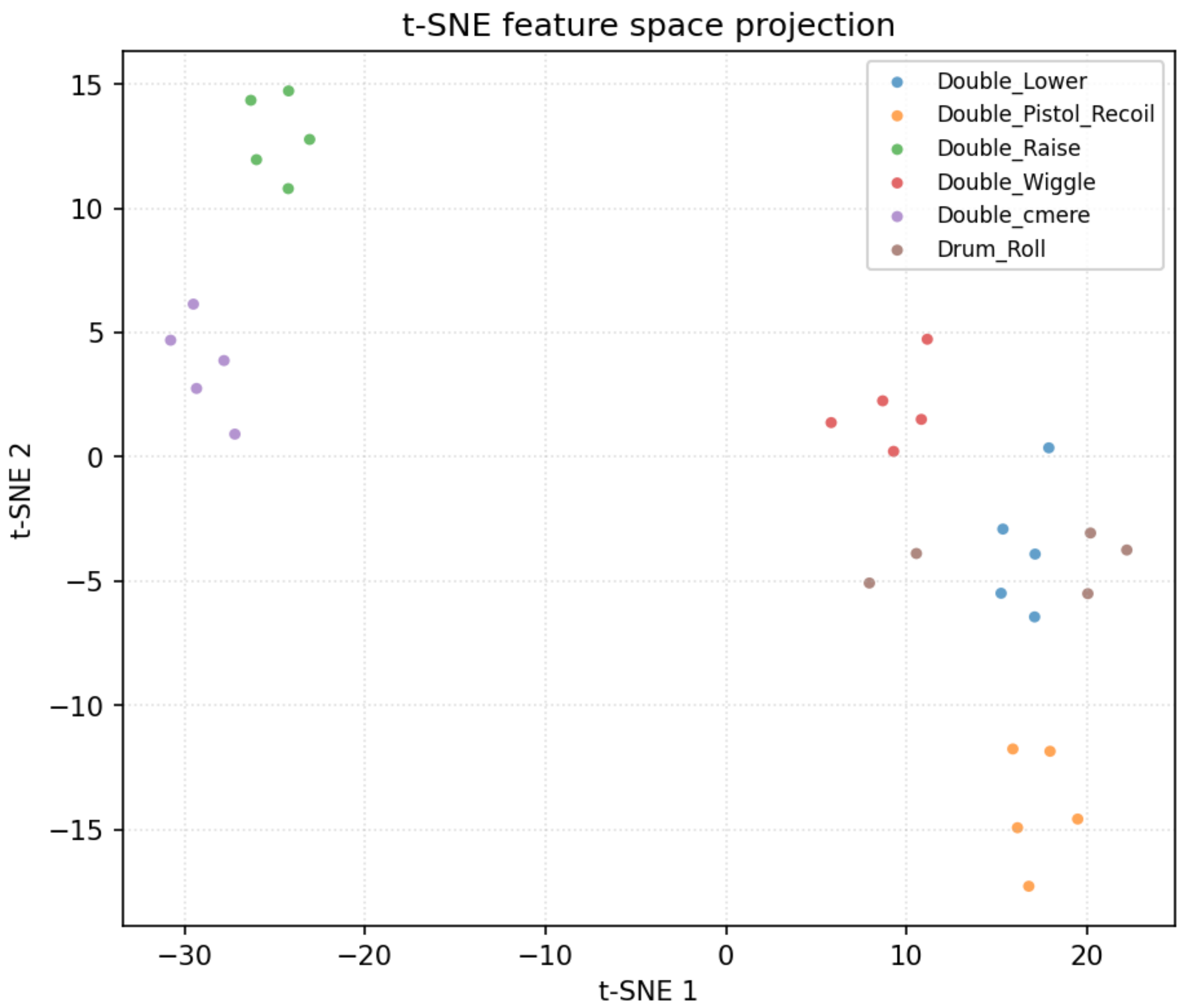
Reading guide: within-class similarity > 0.95 = very consistent; < 0.80 = high variance. Separation ratio > 100x = trivially separable, 20-100x = easy, 5-20x = moderate, < 5x = hard (expect confusion).

Data diagnostics — user 11_Mansh

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

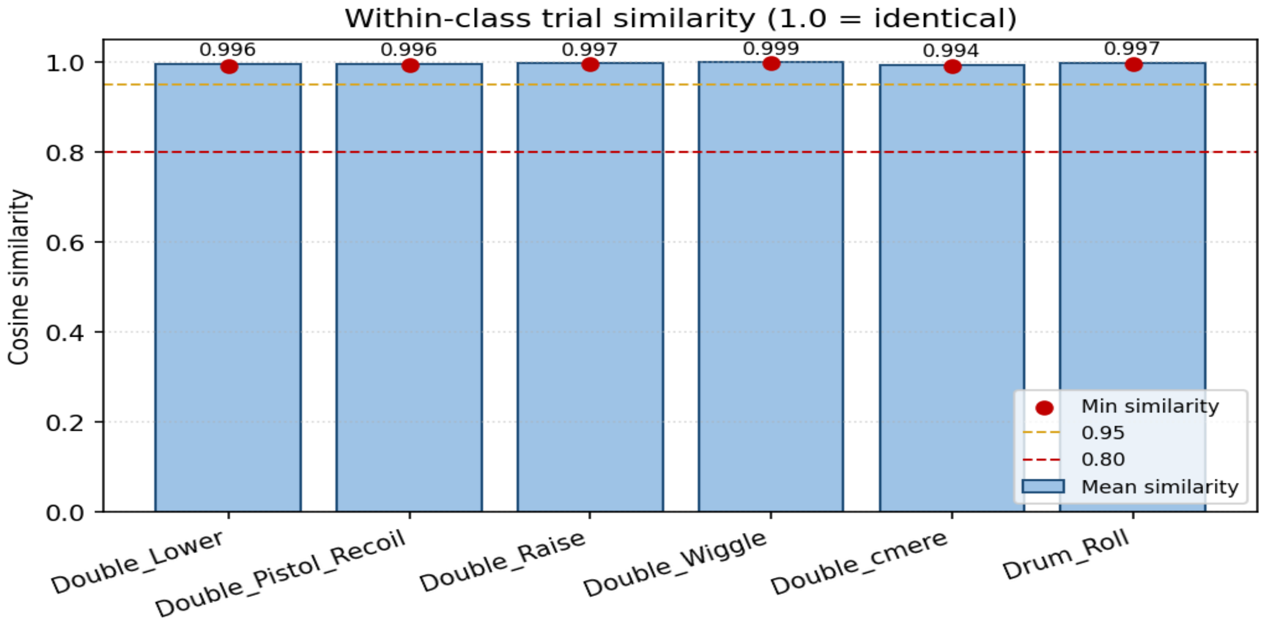


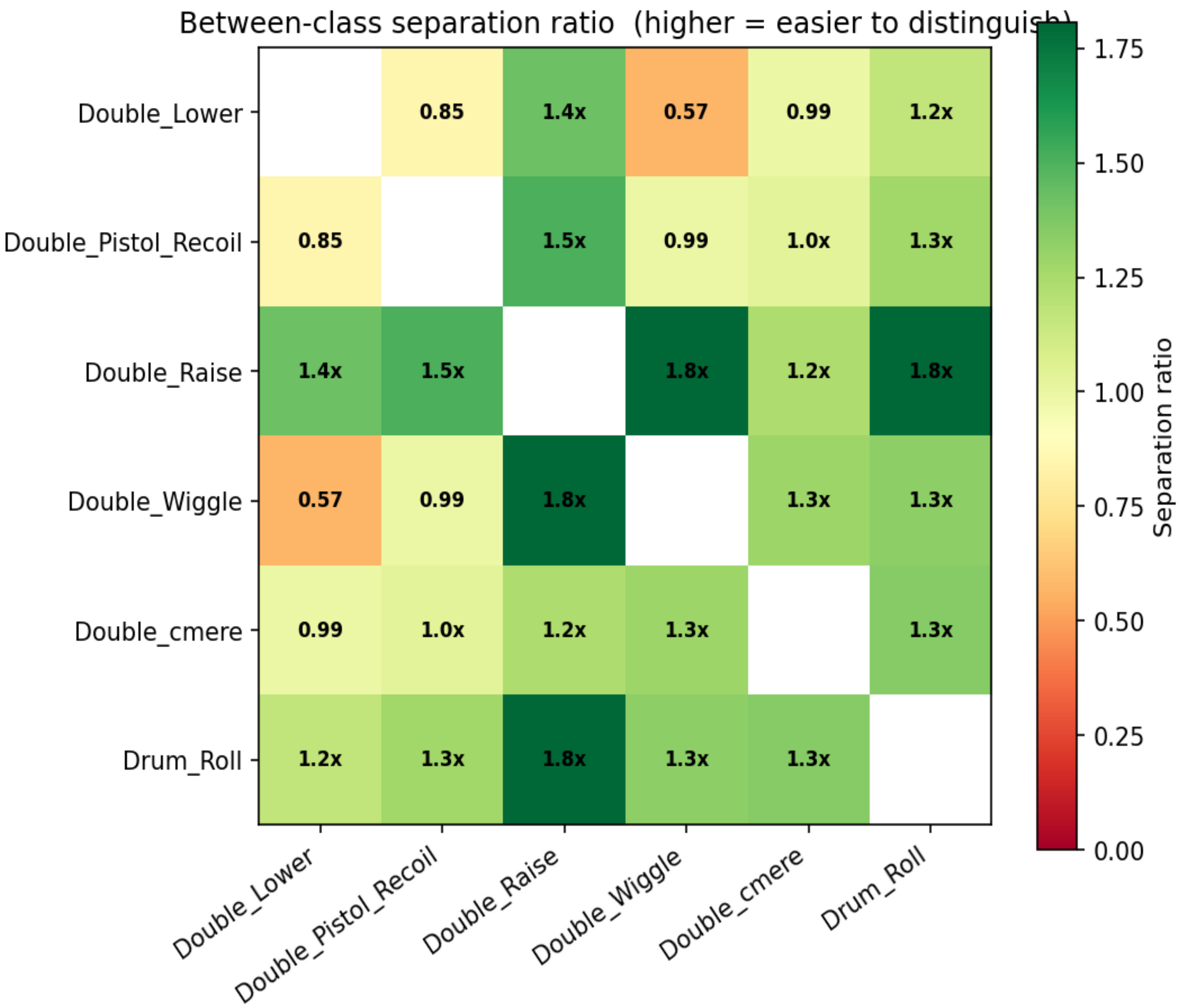


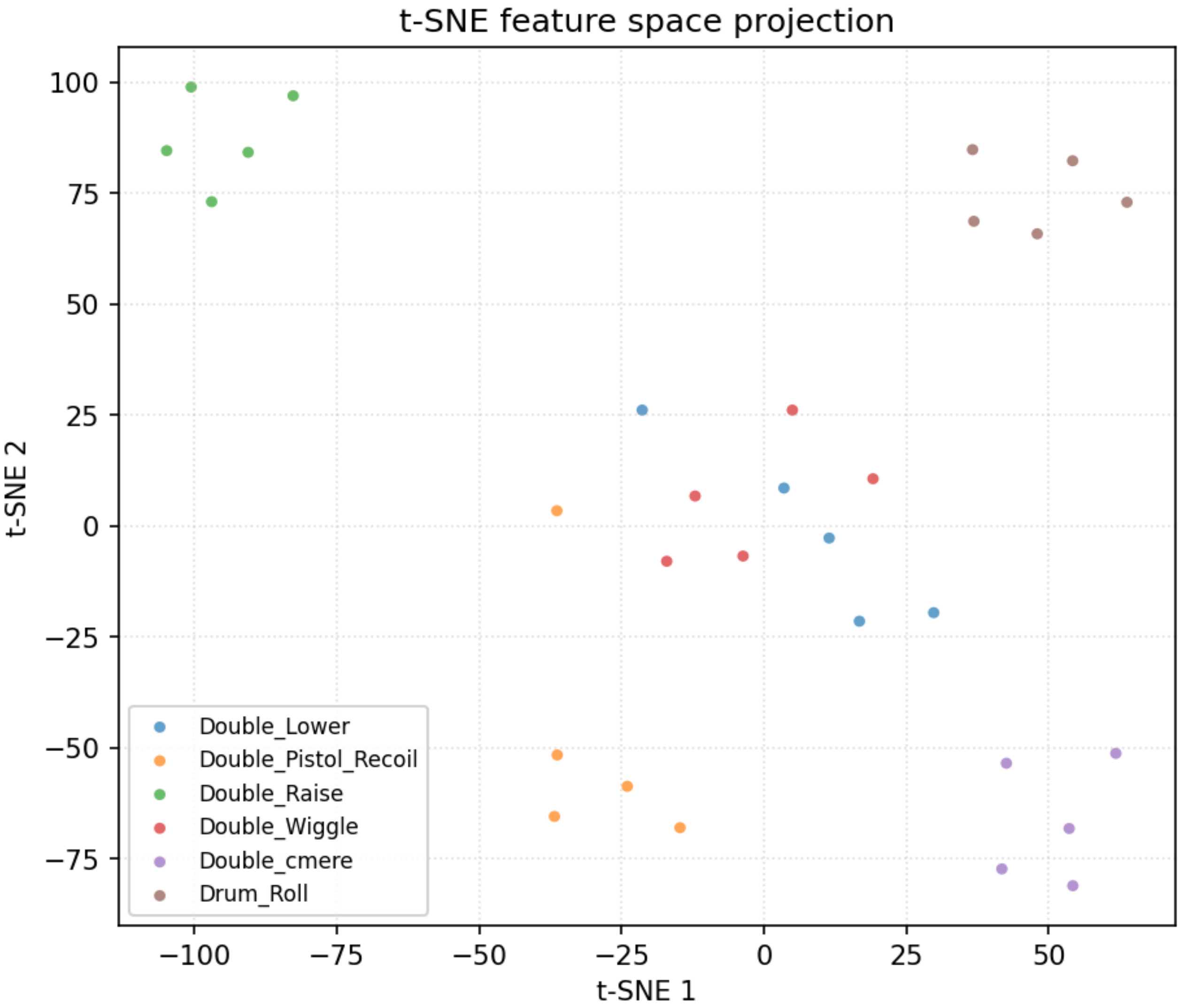

t-SNE feature space projection
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
t-SNE 1
t-SNE 2
15
10
5
0
−5
−10
−15
−30
−20
−10
0
10
20

Data diagnostics — user 12_Marcus

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

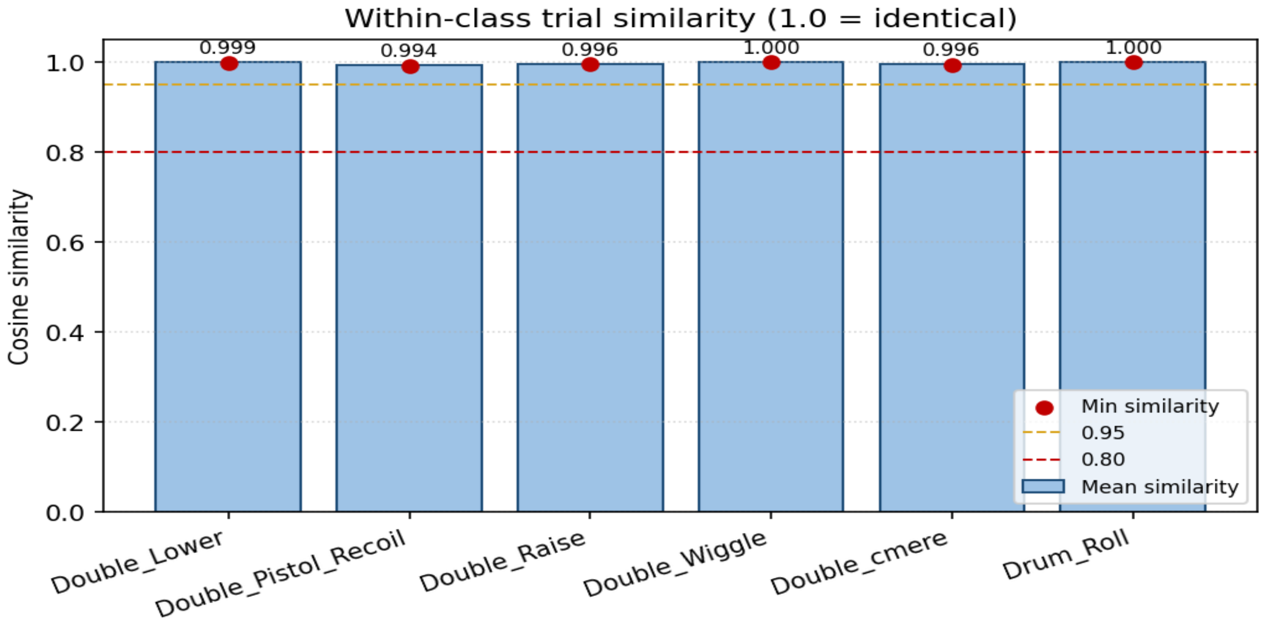


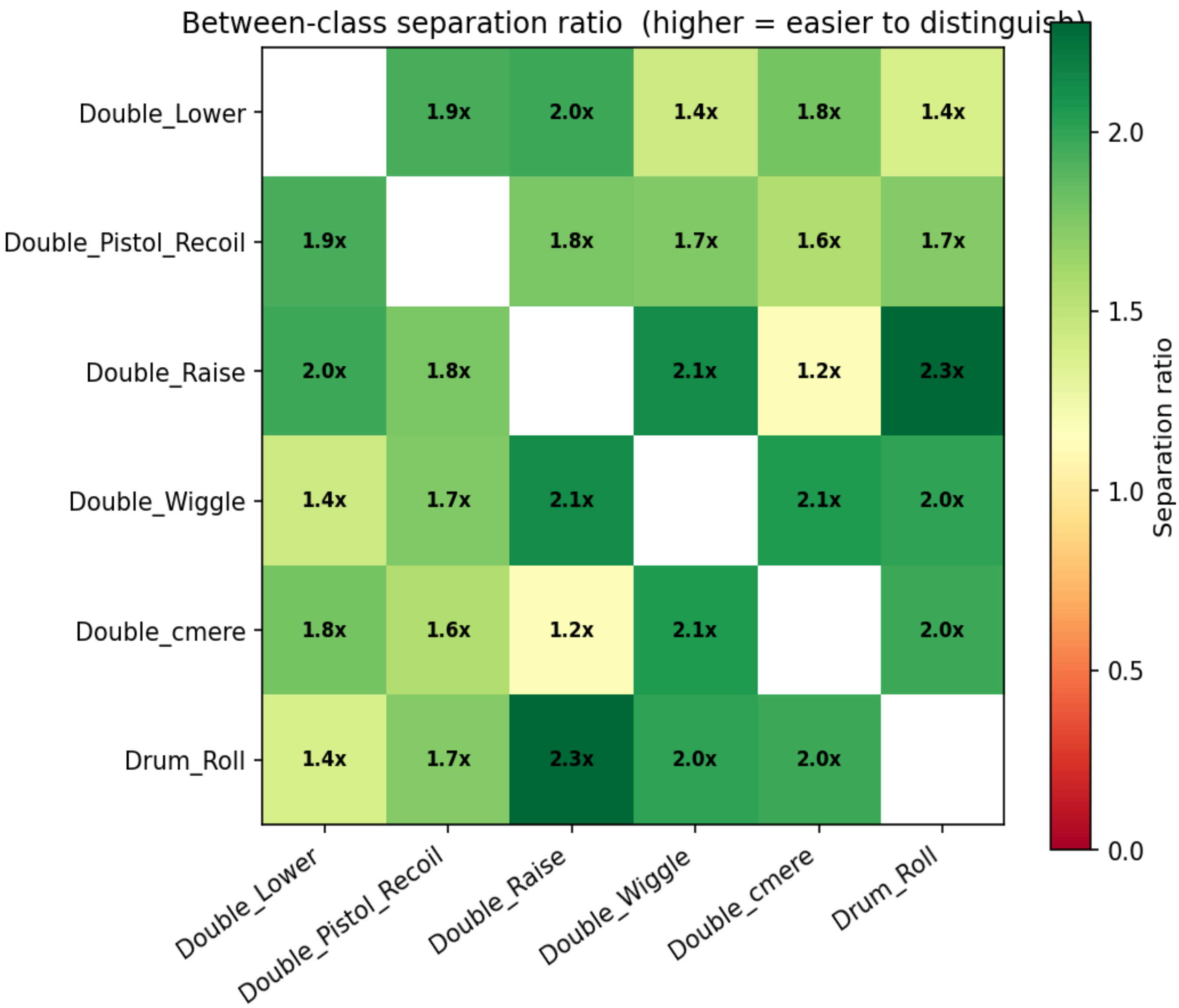


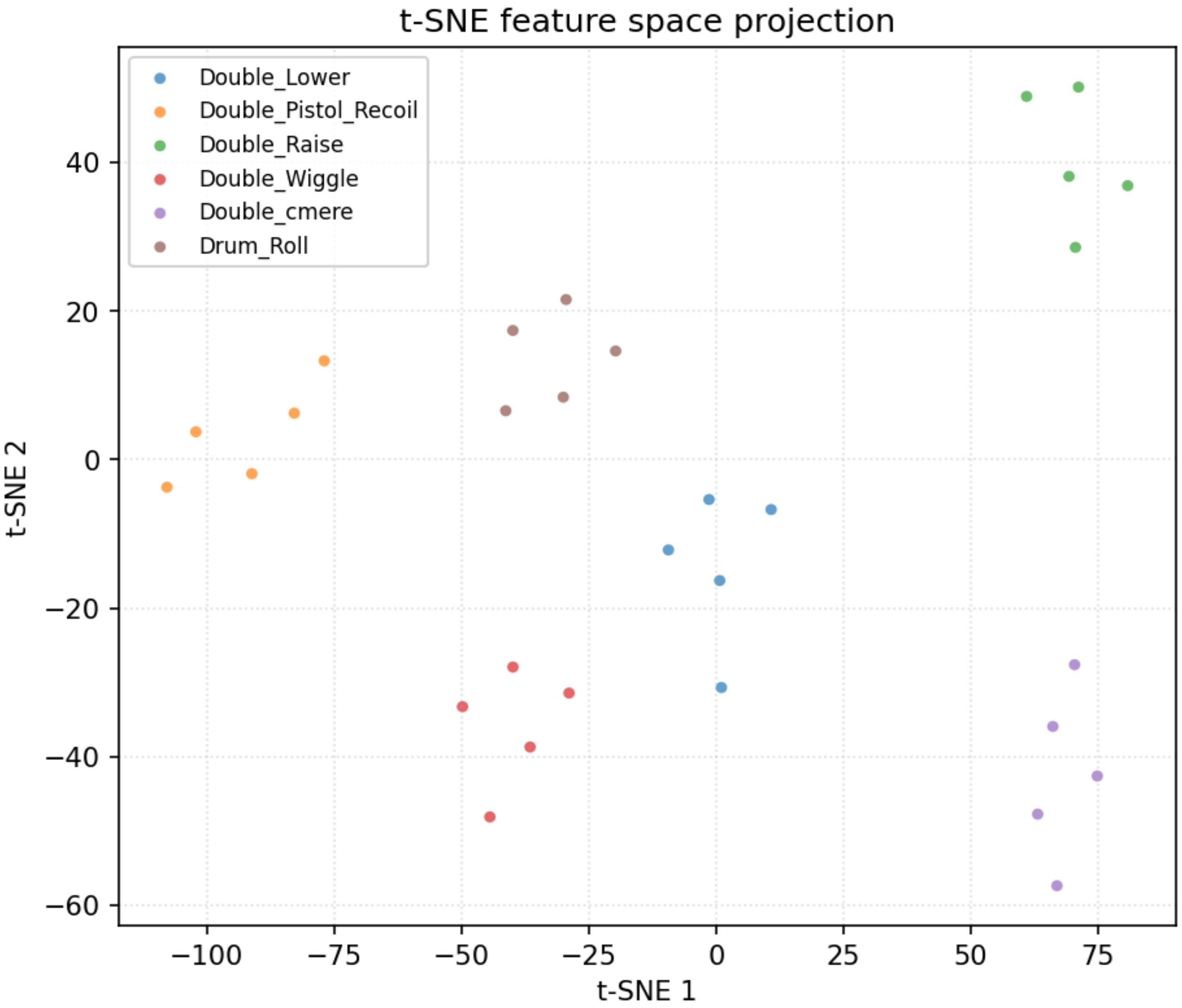

t-SNE feature space projection
t-SNE 2
t-SNE 1
100
75
50
25
0
−25
−50
−75
−100
−75
−50
−25
0
25
50
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll

Data diagnostics — user 14_StephenV2

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

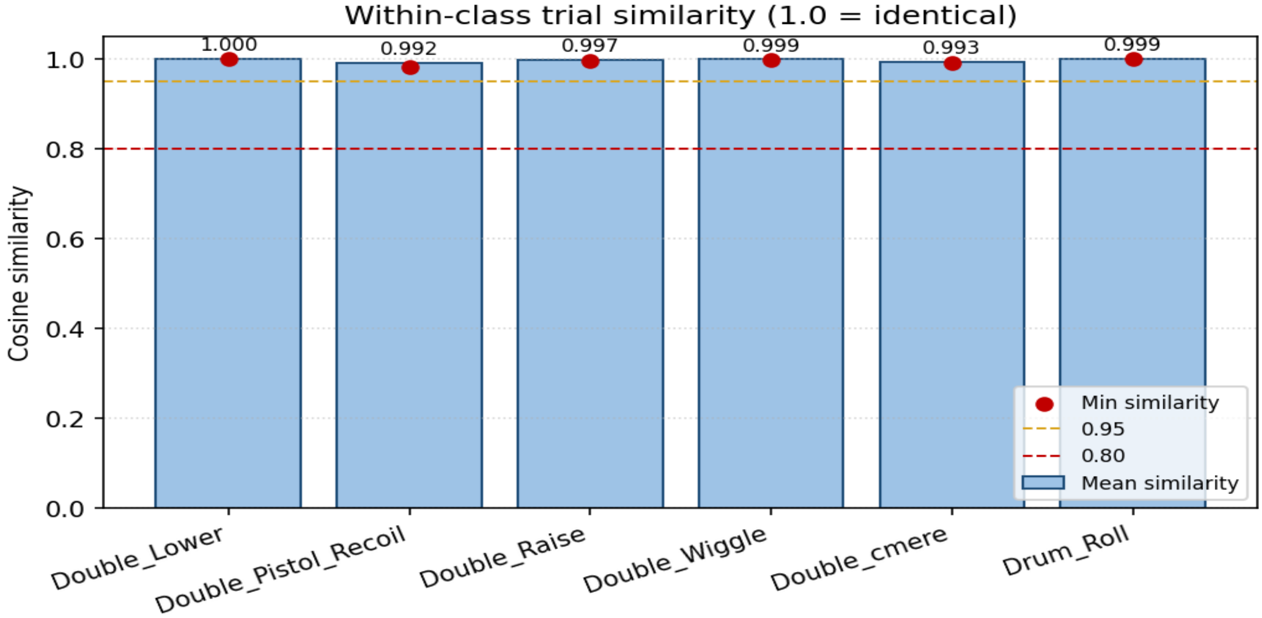


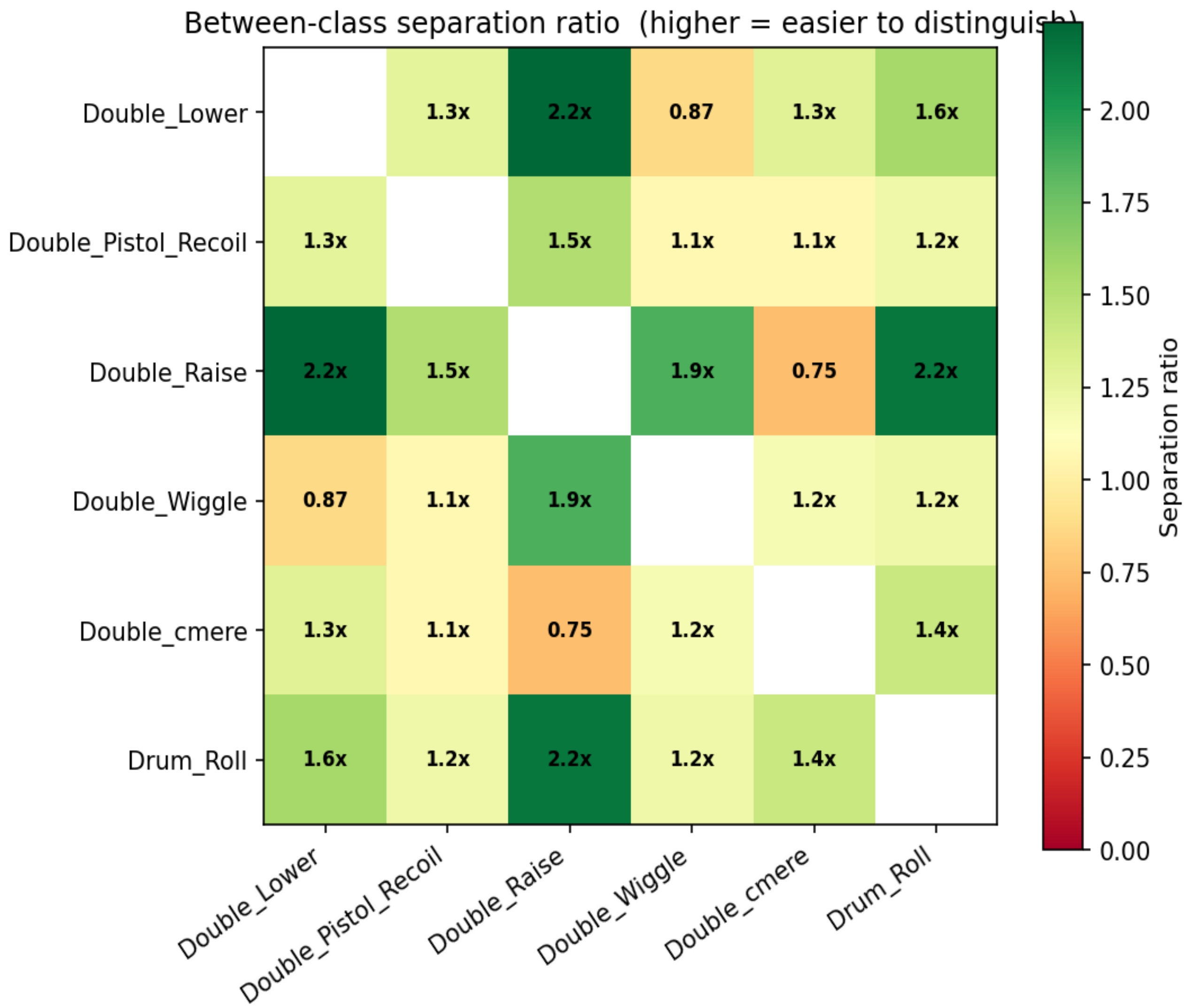


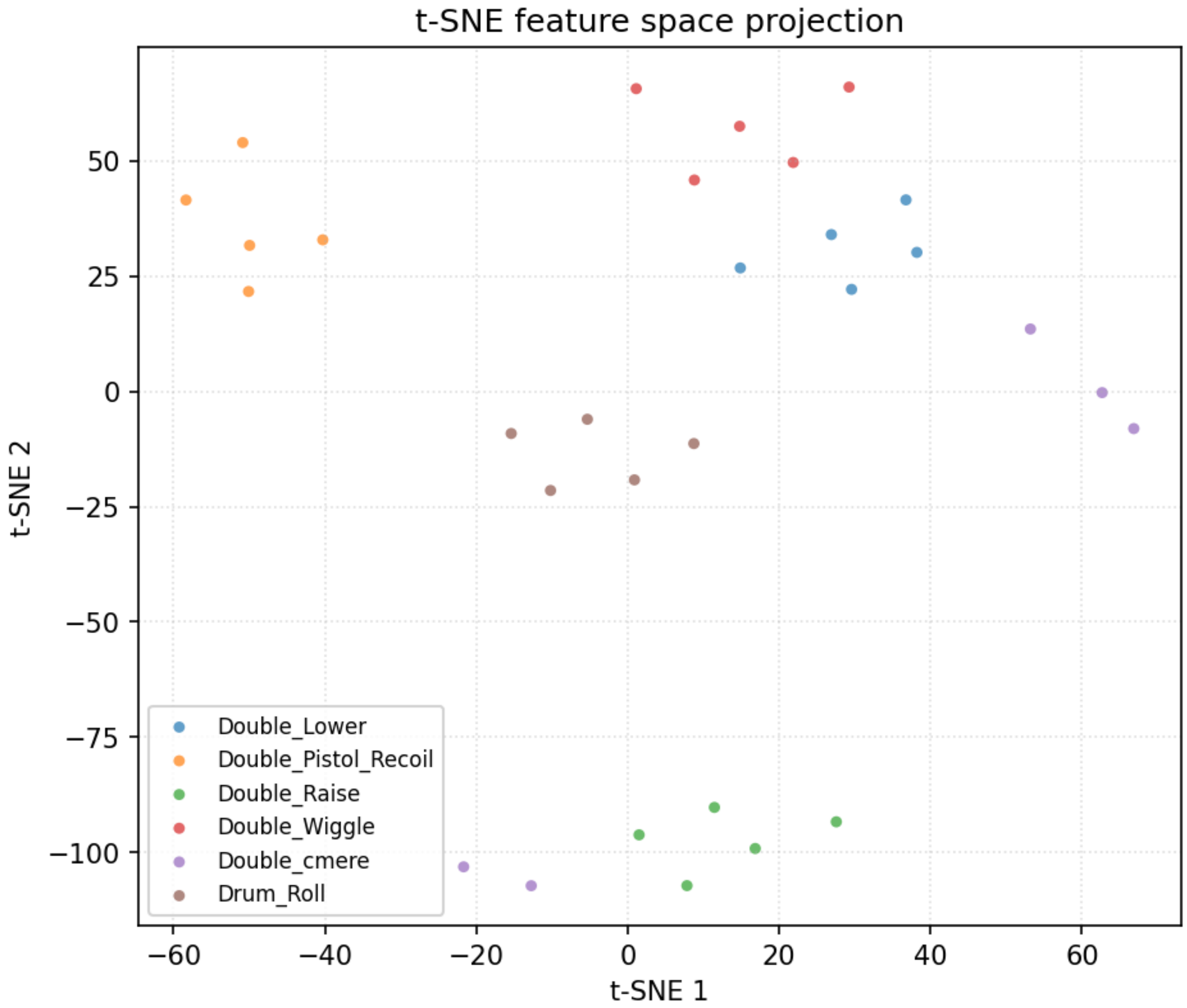

t-SNE feature space projection
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
40
20
0
−20
−40
−60
t-SNE 2
−100
−75
−50
−25
0
25
50
75
t-SNE 1

Data diagnostics — user 15_Tash

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

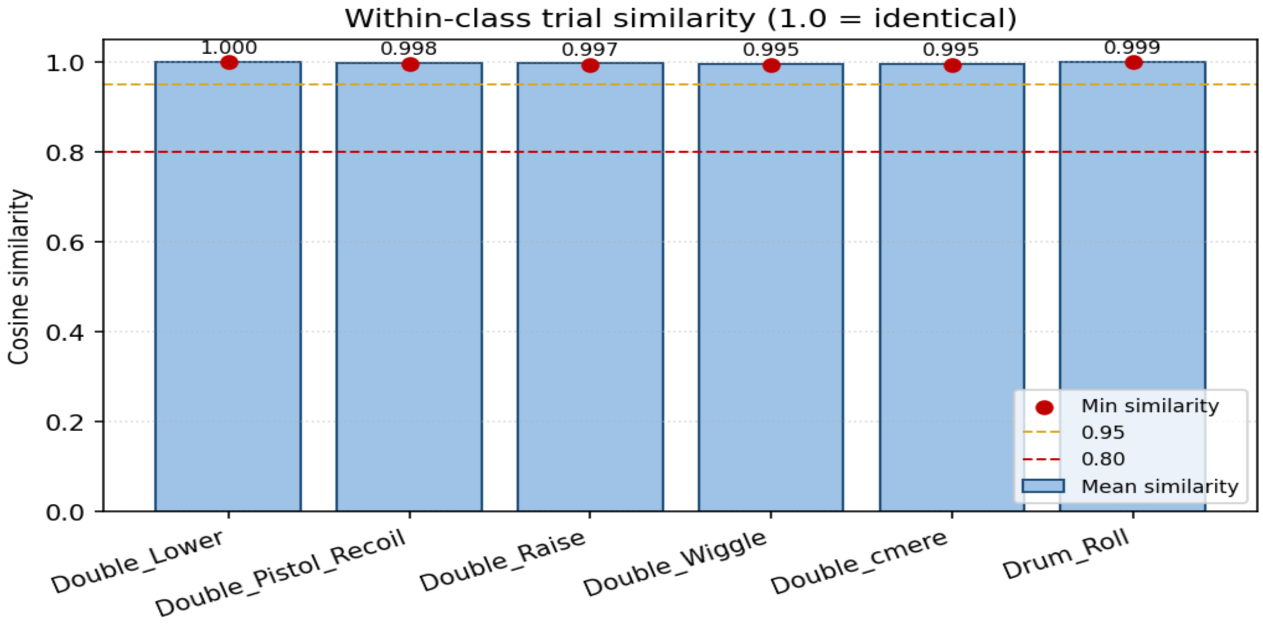


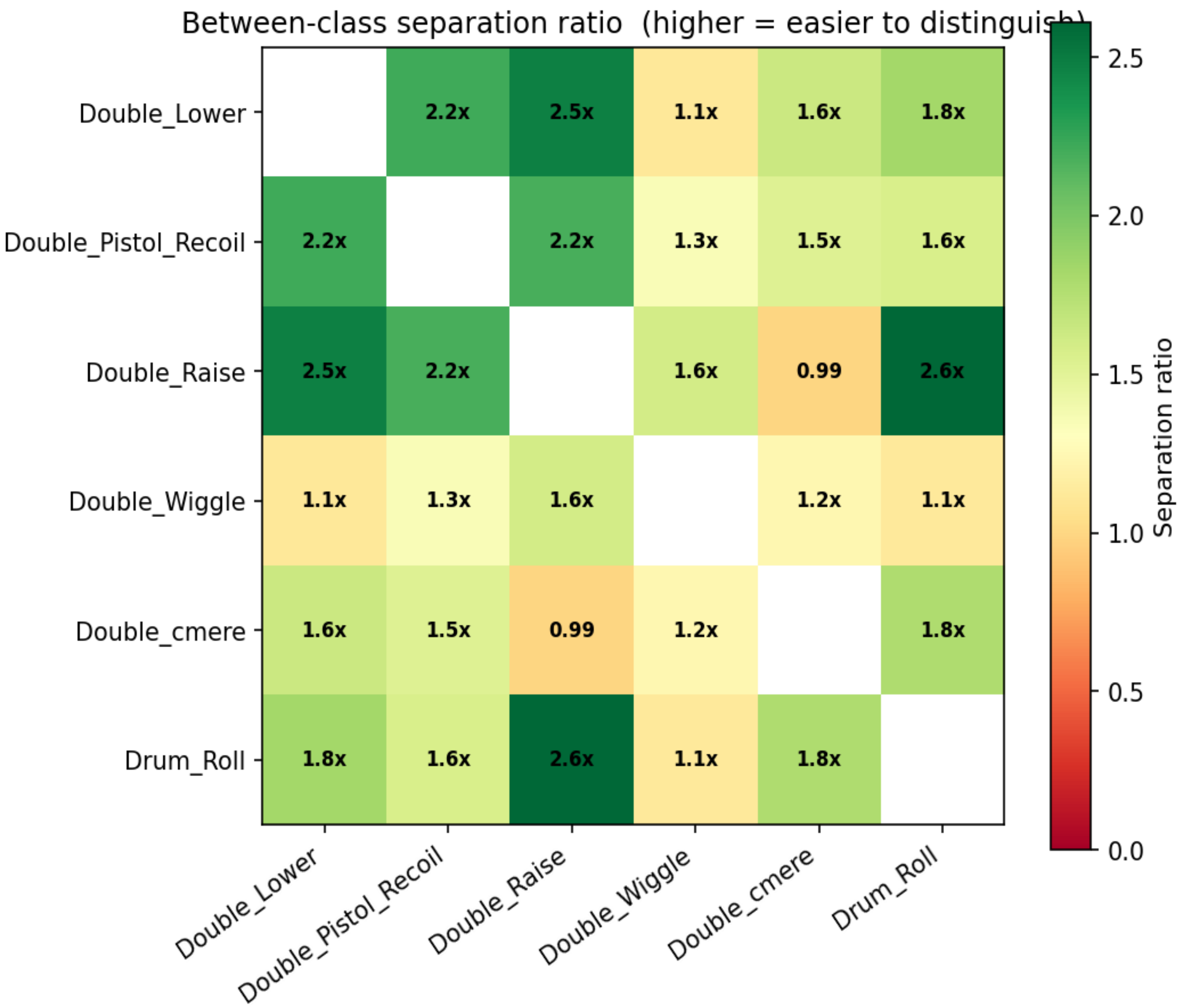


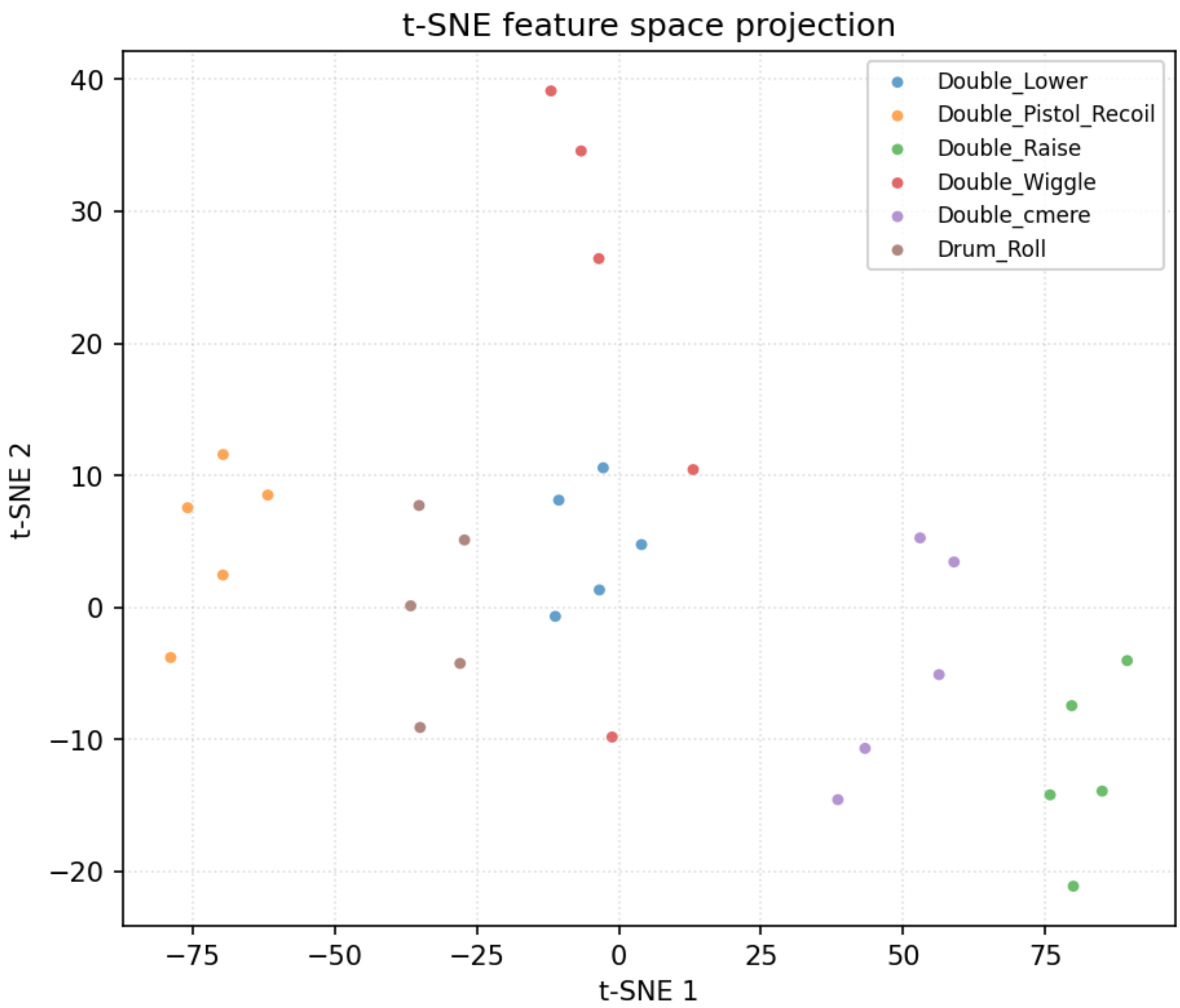

t-SNE feature space projection
t-SNE 2
t-SNE 1
50
25
0
−25
−50
−75
−100
−60
−40
−20
0
20
40
60
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll

Data diagnostics — user 16_Bella

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

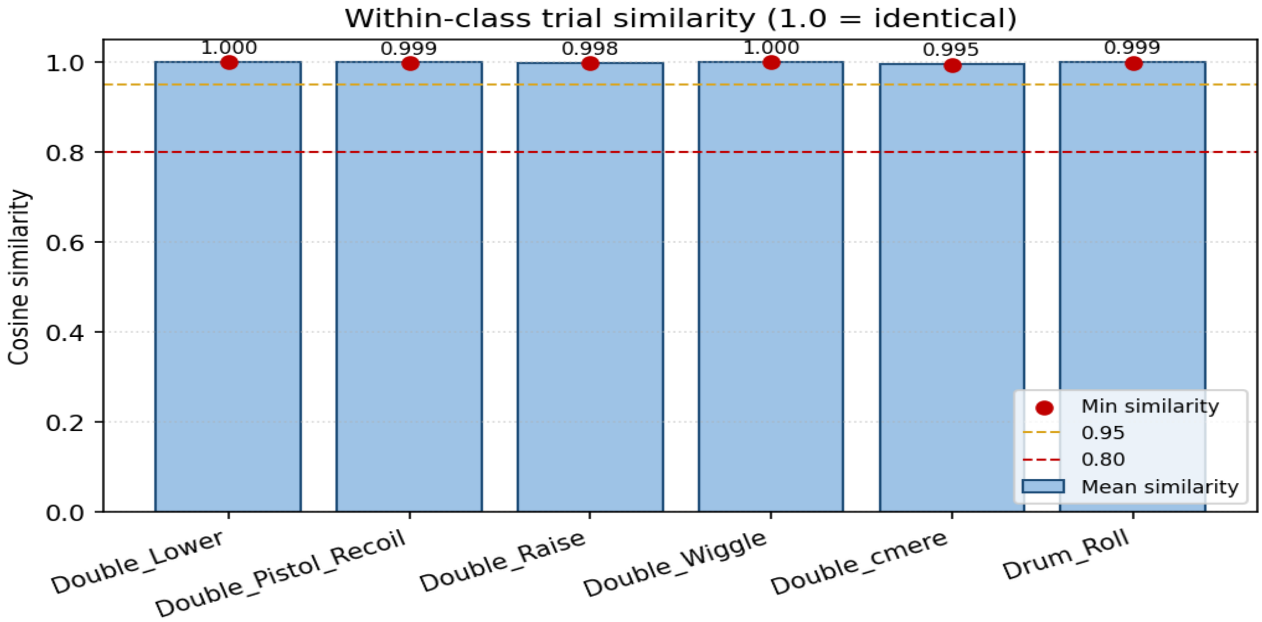


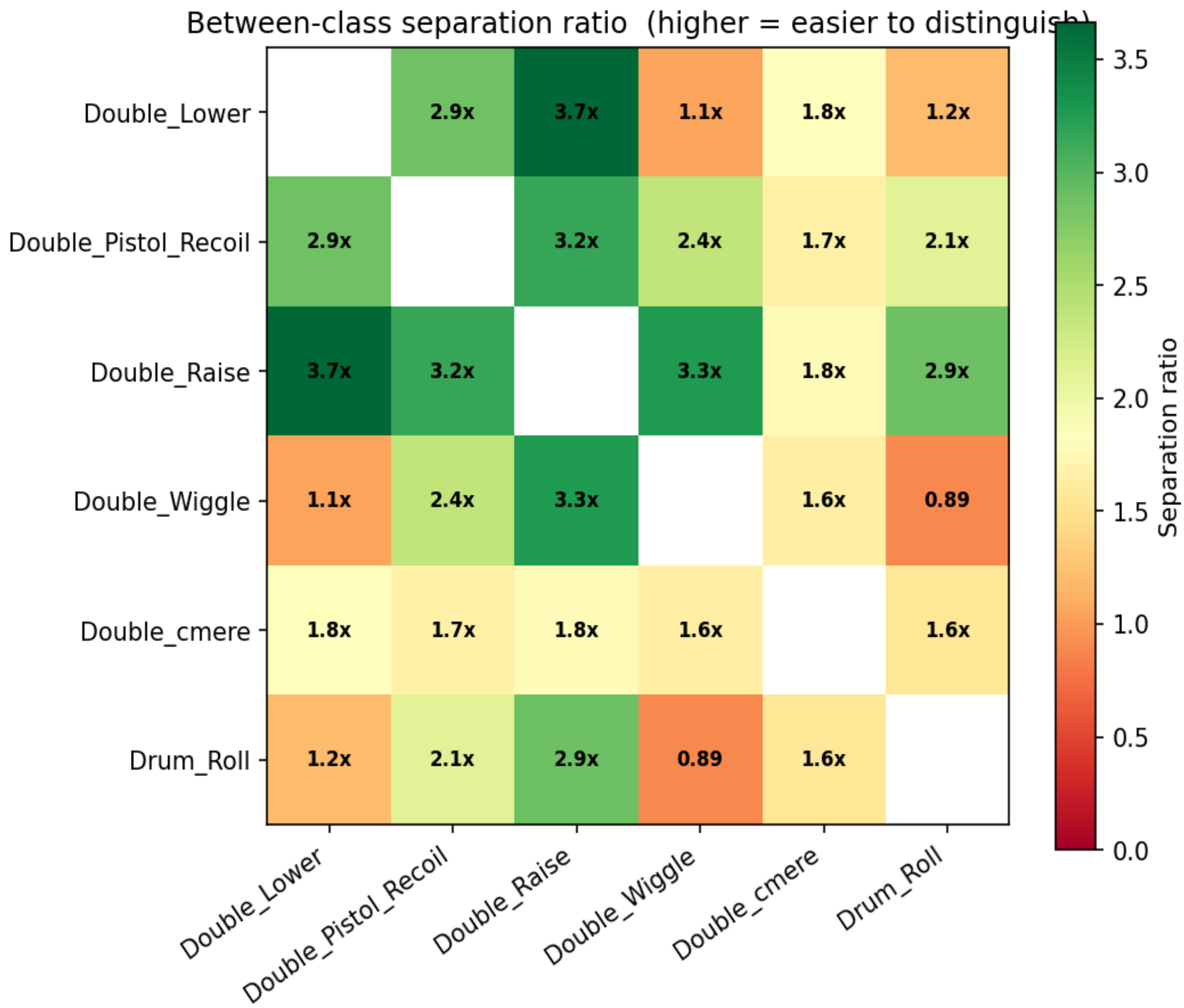


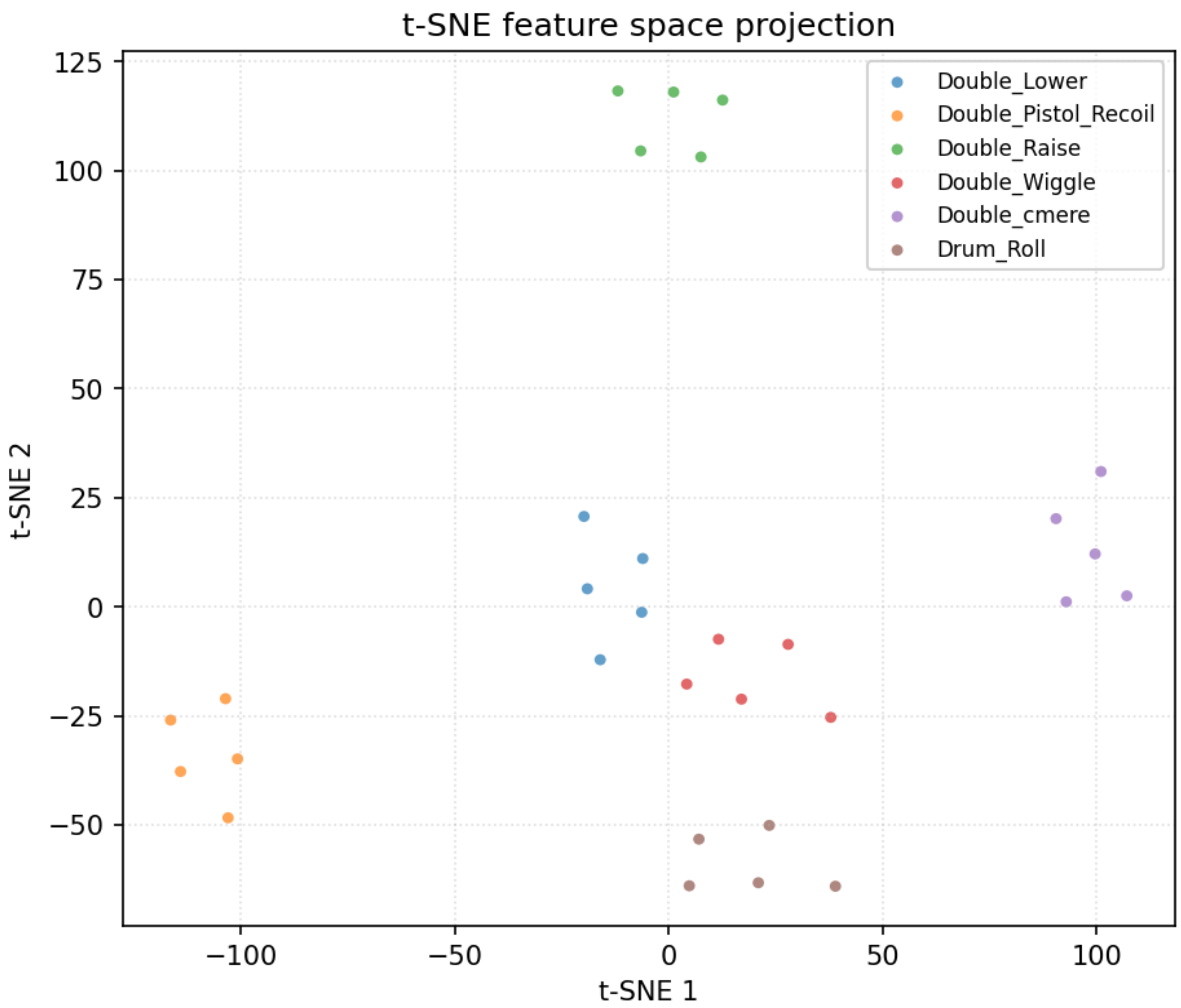

t-SNE feature space projection
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
t-SNE 2
t-SNE 1
40
30
20
10
0
−10
−20
−75
−50
−25
0
25
50
75

Data diagnostics — user 17_Raquel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

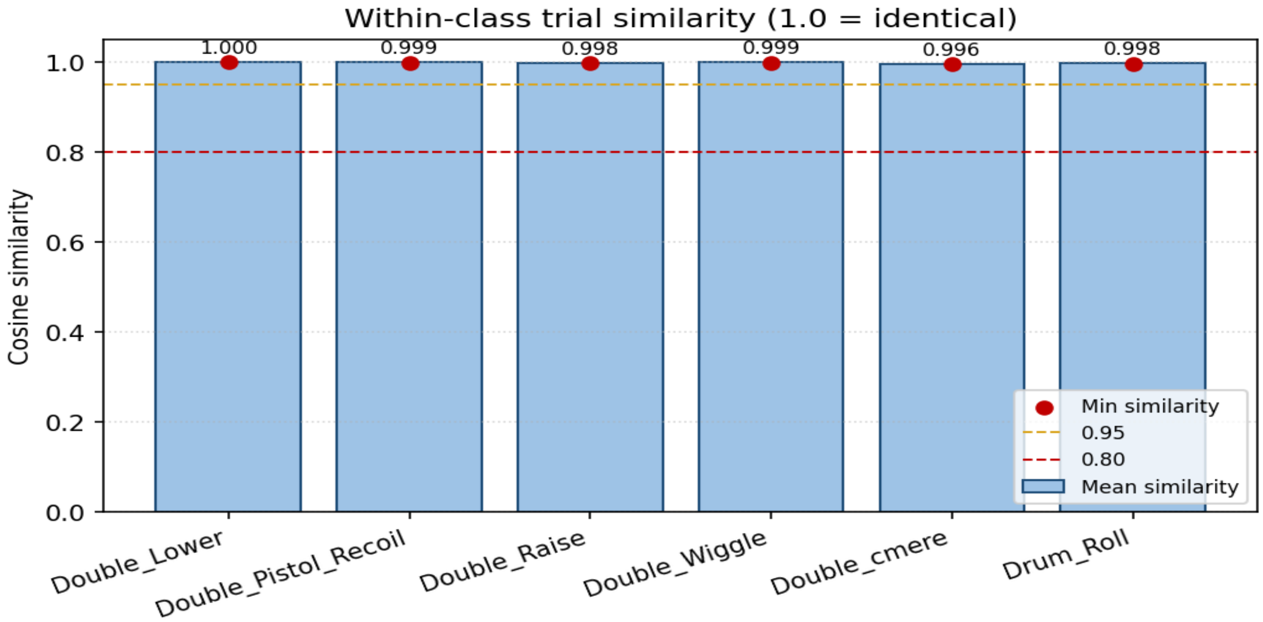


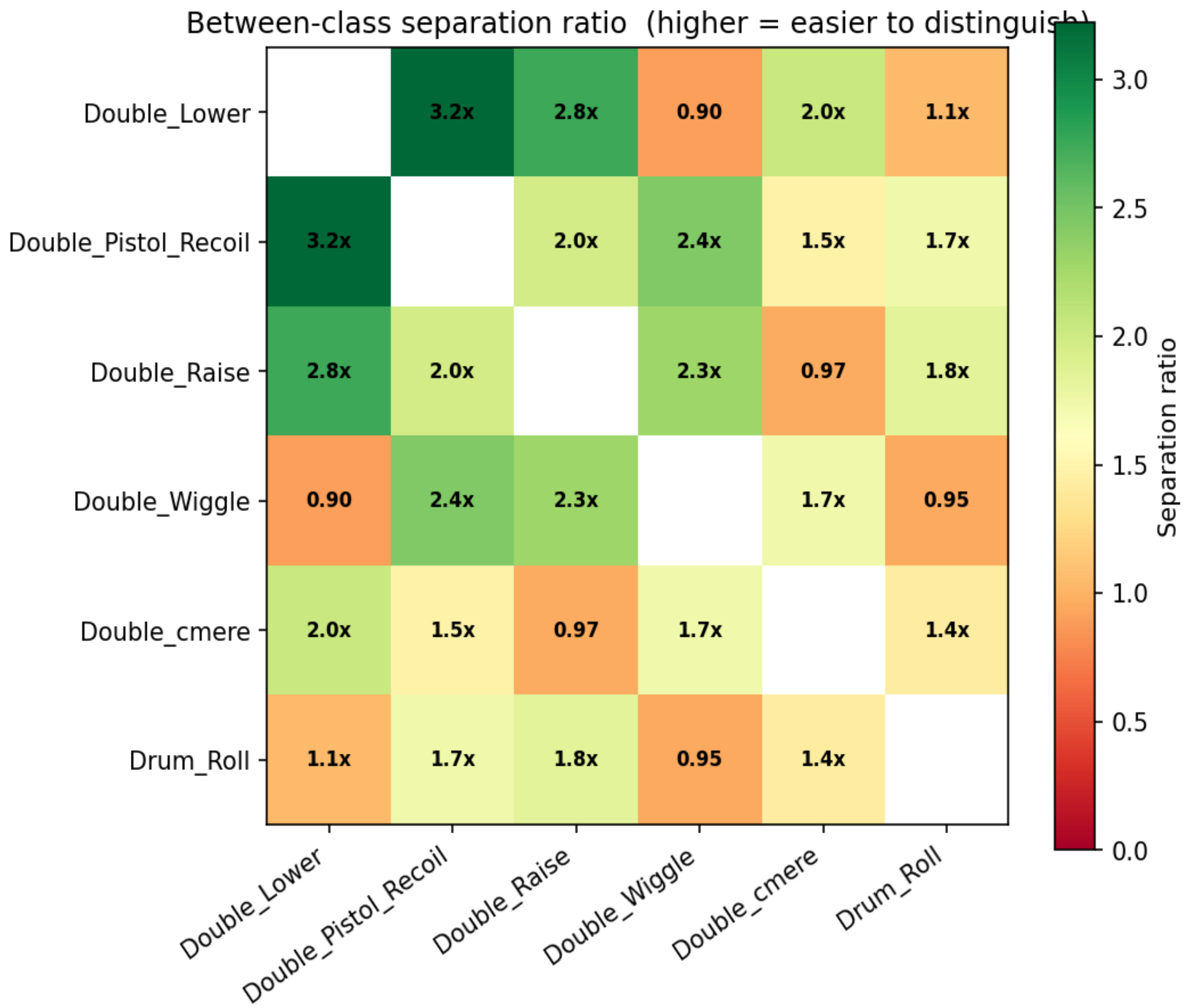



t-SNE feature space projection
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
t-SNE 2
t-SNE 1
125
100
75
50
25
0
−25
−50
−100
−50
0
50
100

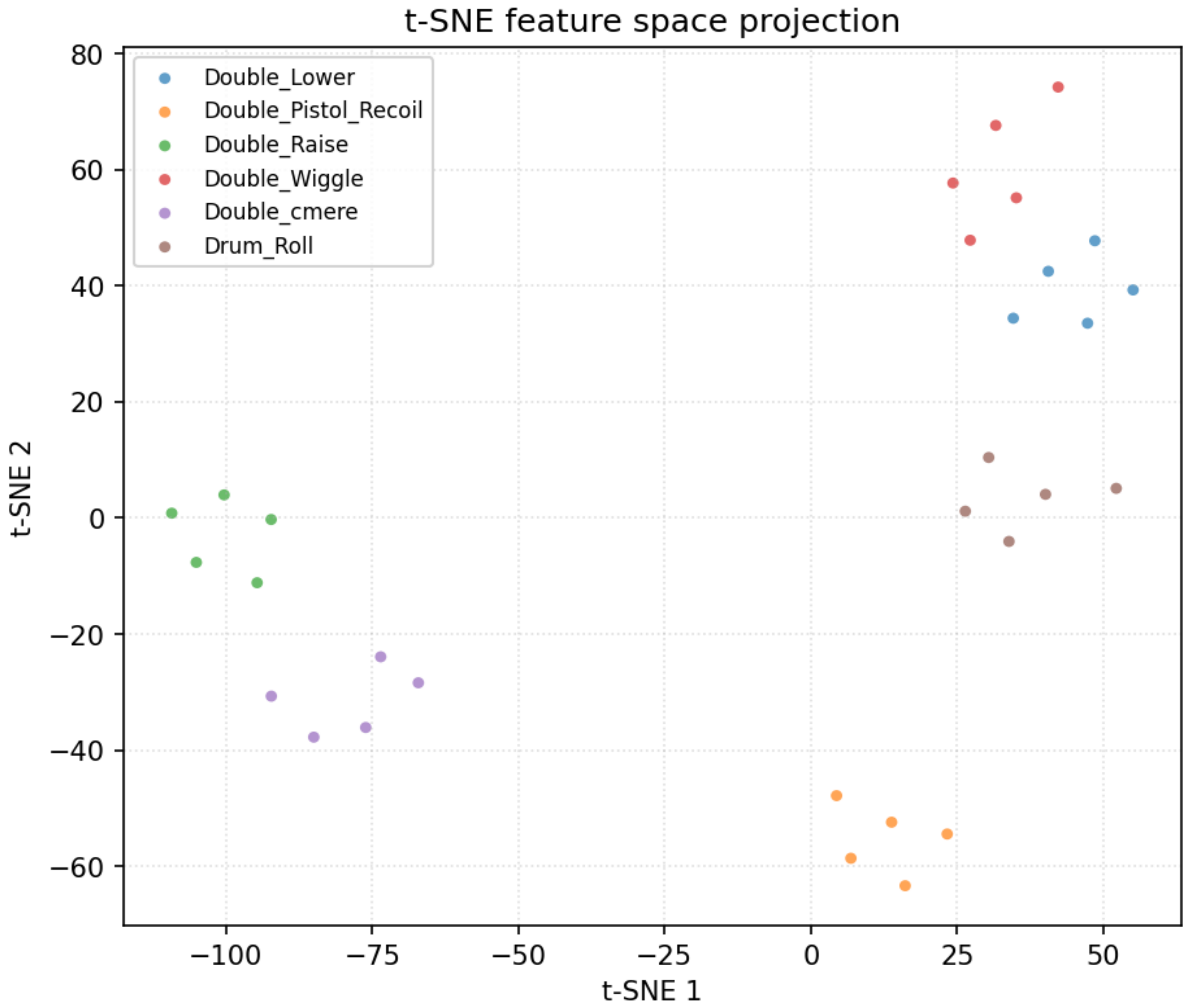
Data diagnostics — user 18_Bridgette

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).



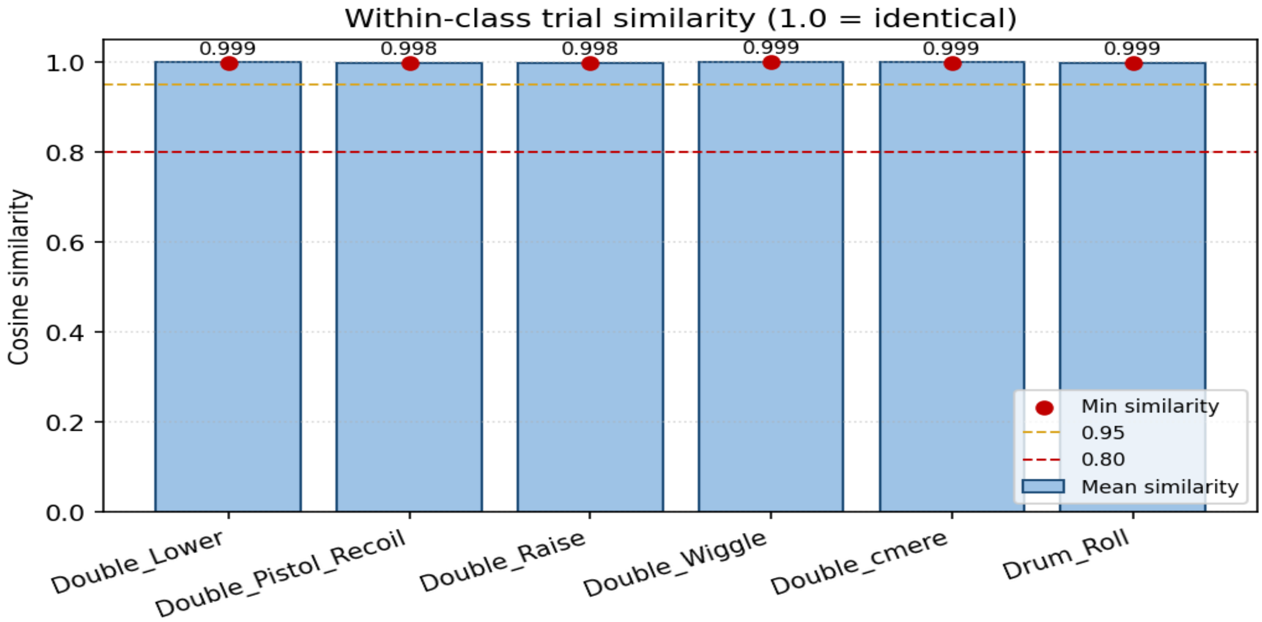


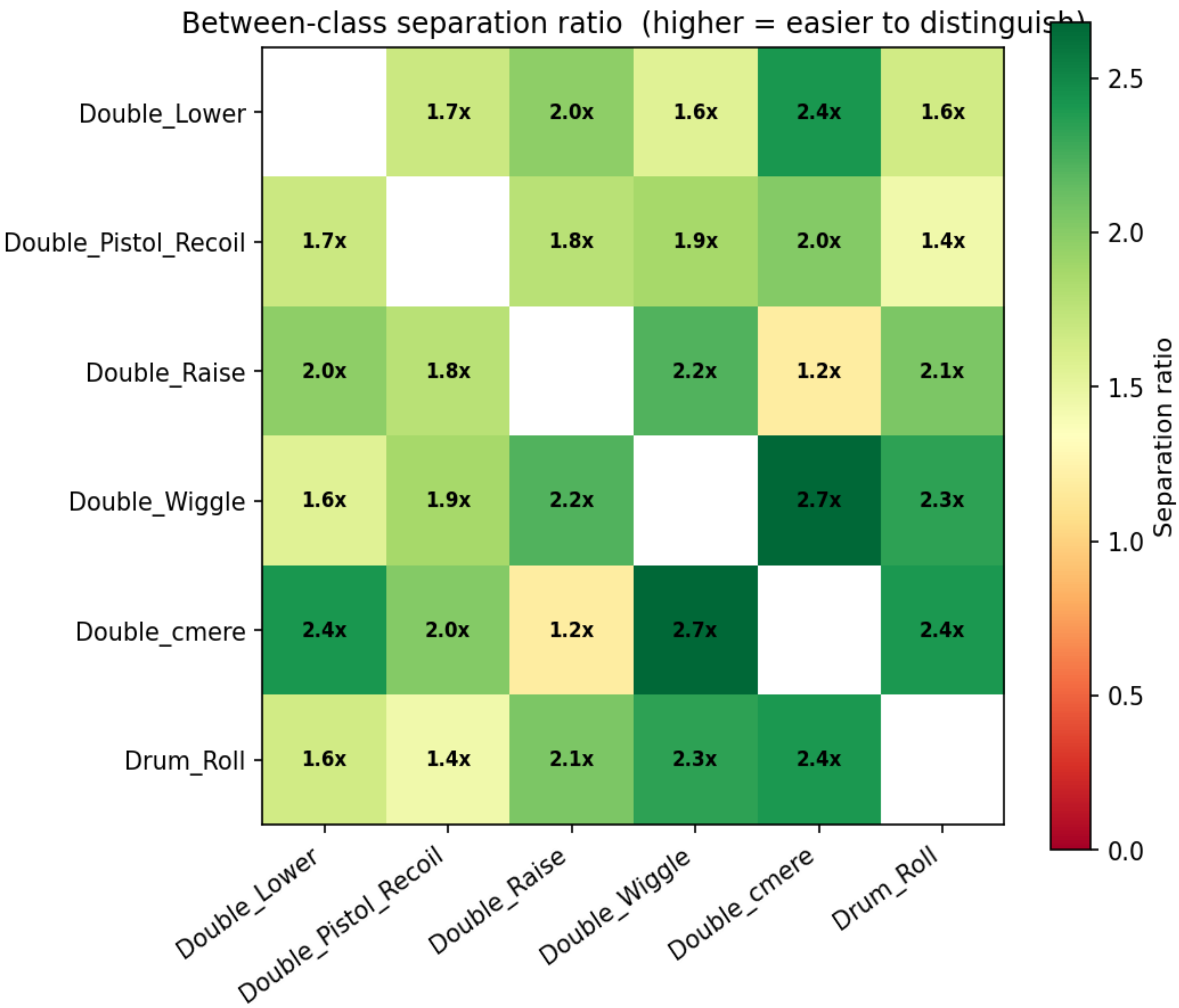
	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll
Double_Lower		3.2x	2.8x	0.90	2.0x	1.1x
Double_Pistol_Recoil	3.2x		2.0x	2.4x	1.5x	1.7x
Double_Raise	2.8x	2.0x		2.3x	0.97	1.8x
Double_Wiggle	0.90	2.4x	2.3x		1.7x	0.95
Double_cmere	2.0x	1.5x	0.97	1.7x		1.4x
Drum_Roll	1.1x	1.7x	1.8x	0.95	1.4x	

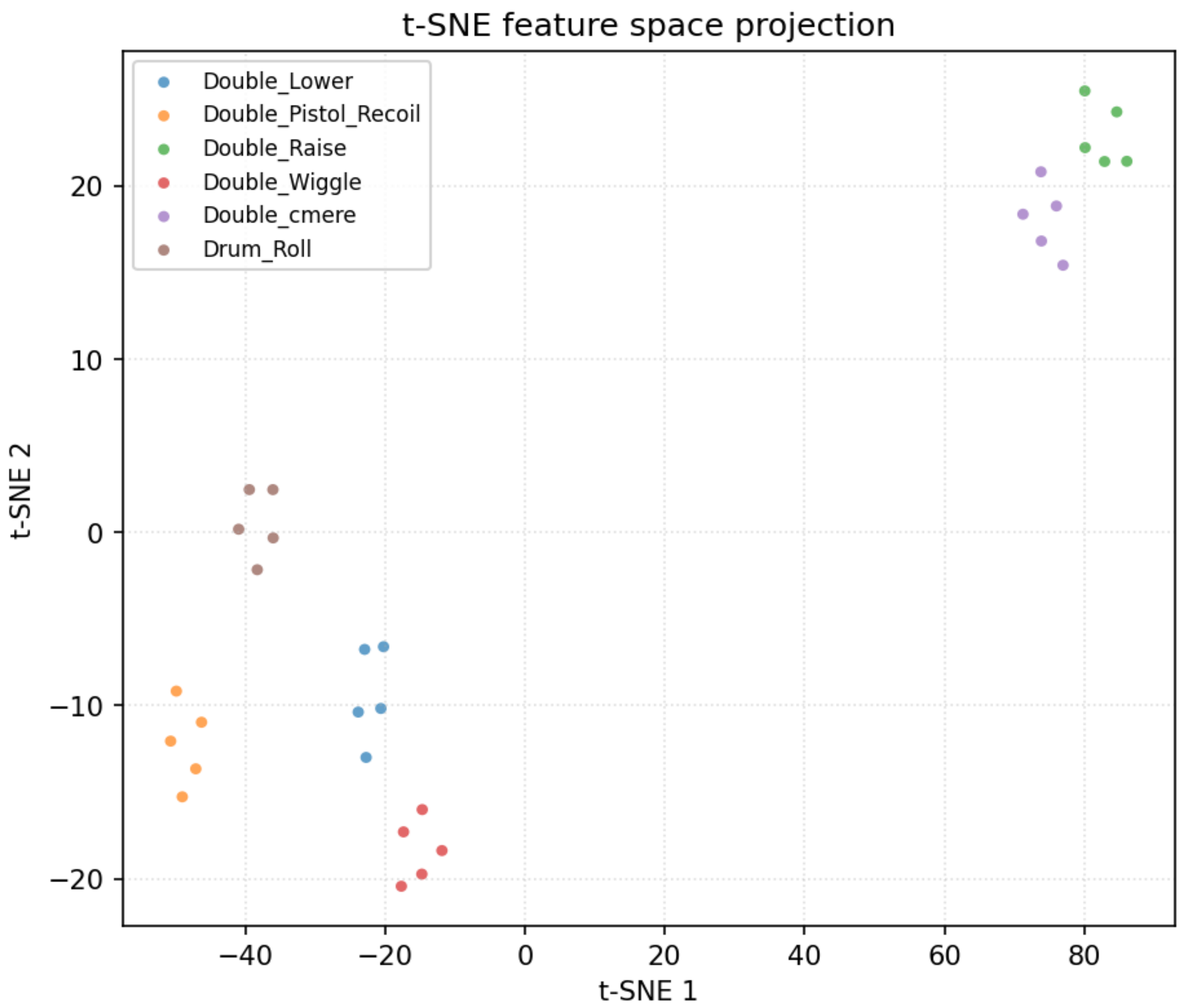

t-SNE feature space projection
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
80
60
40
20
0
−20
−40
−60
−100
−75
−50
−25
0
25
50
t-SNE 1
t-SNE 2

Data diagnostics — user 1_Alan

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

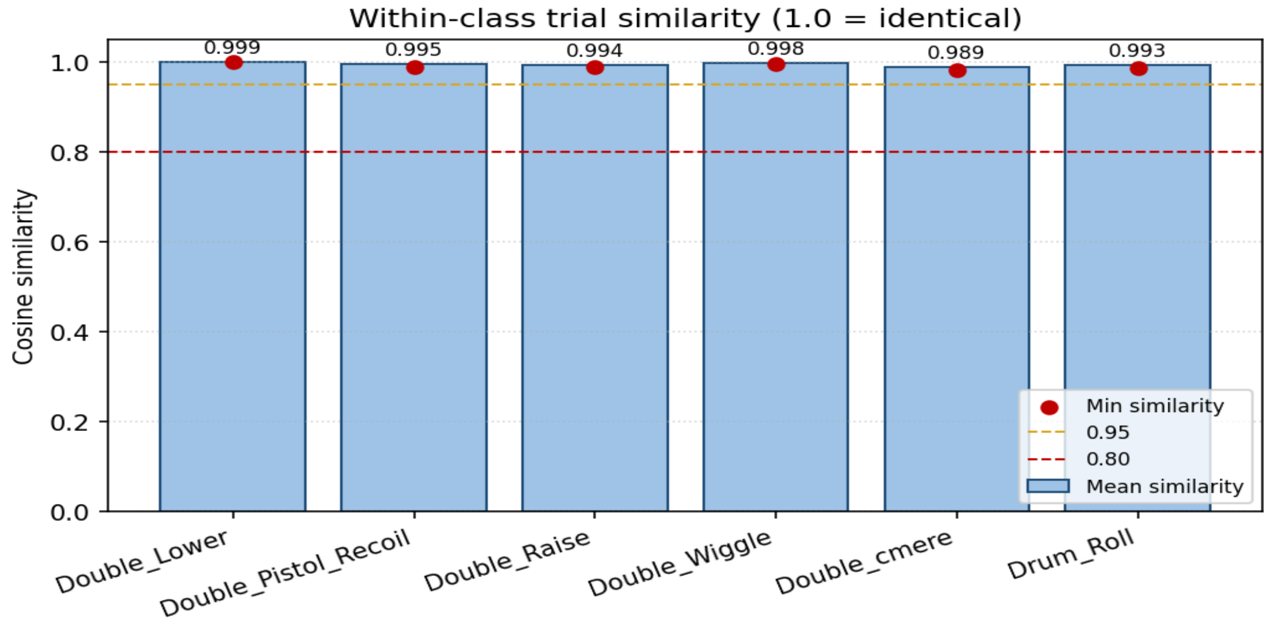


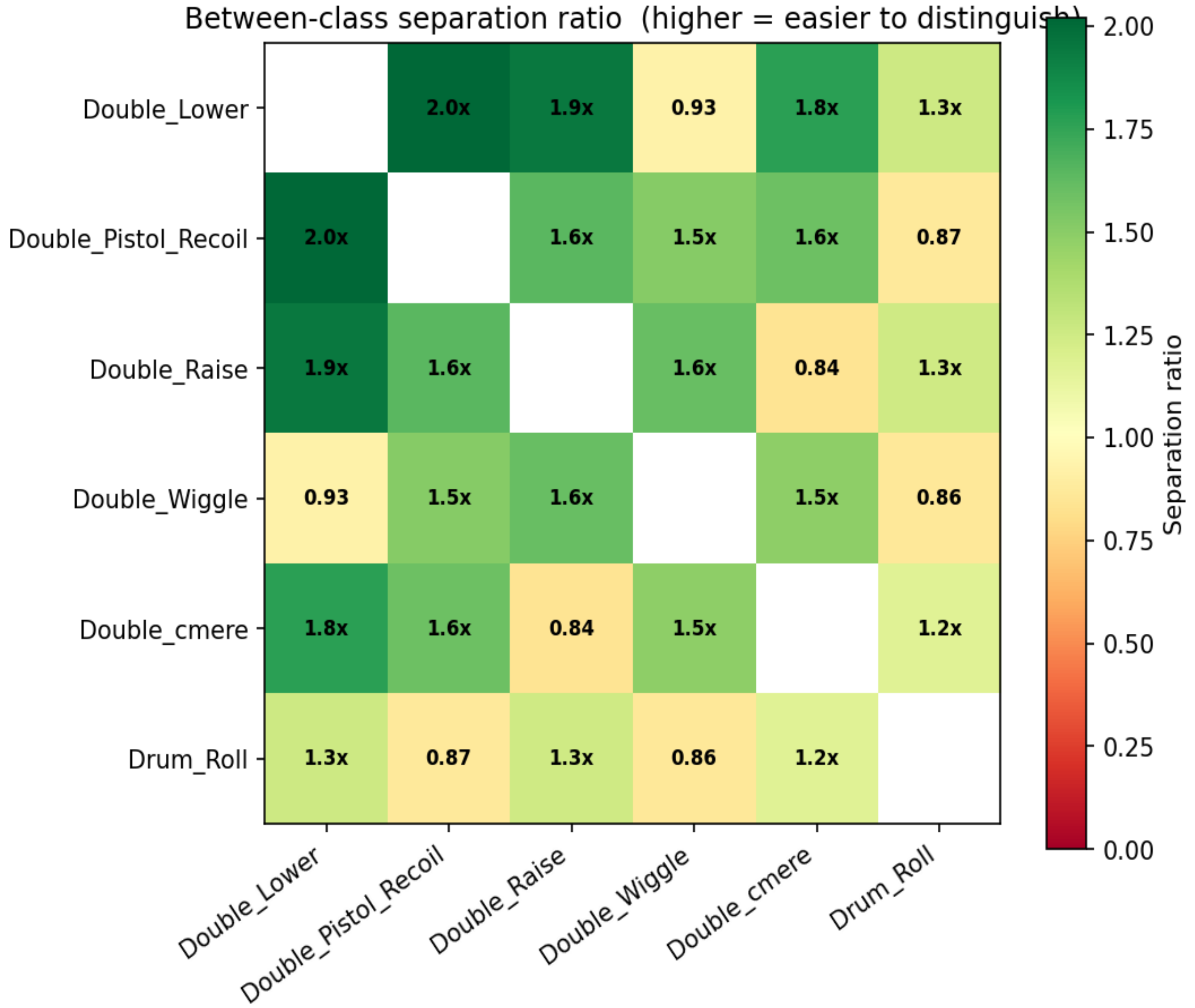


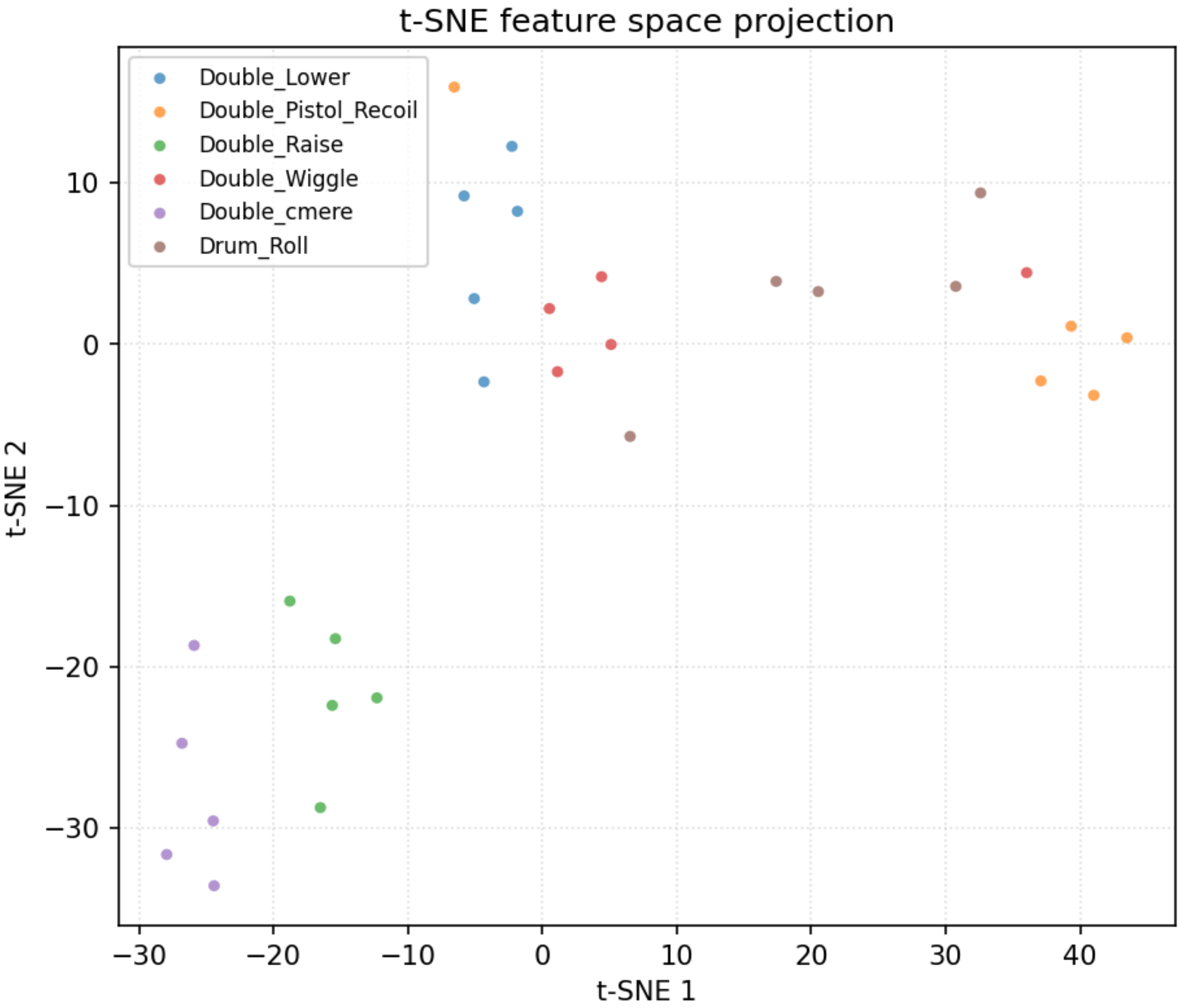

t-SNE feature space projection
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
20
10
0
−10
−20
−40
−20
0
20
40
60
80
t-SNE 1
t-SNE 2

Data diagnostics — user 2_Alex

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

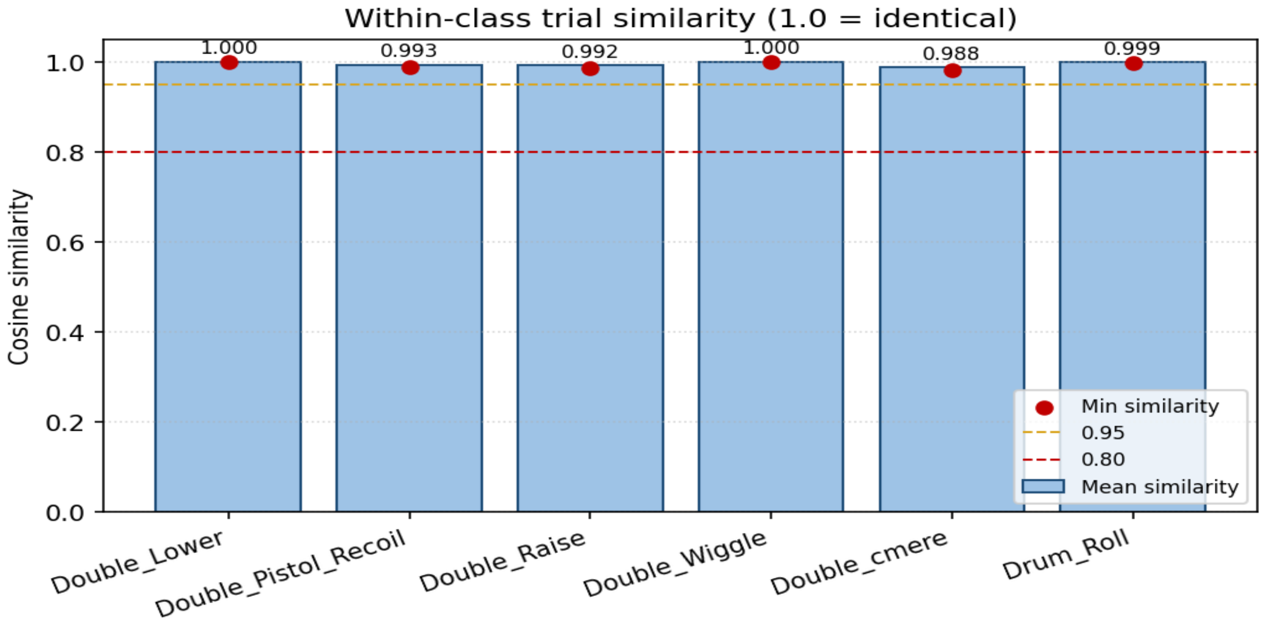


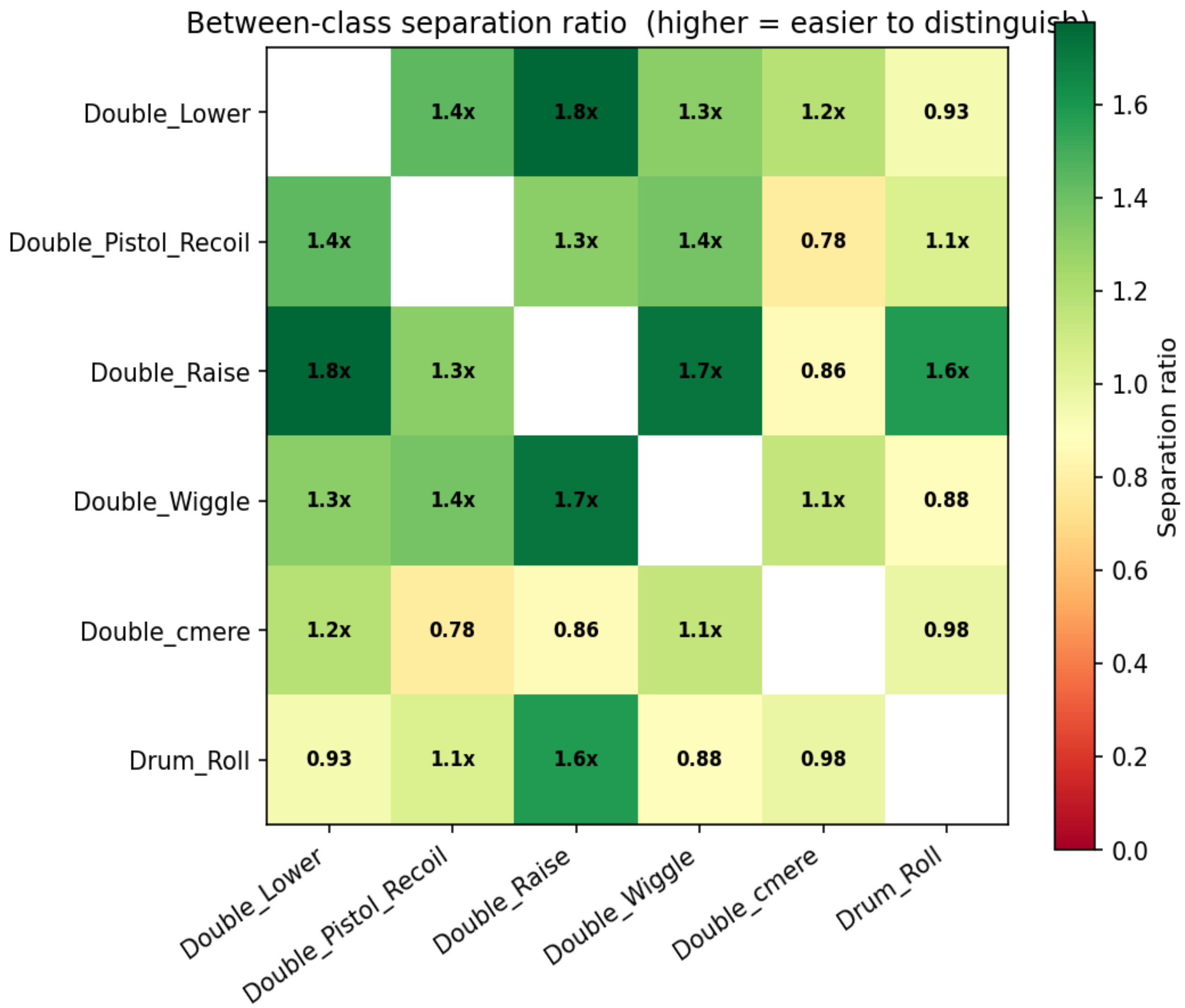


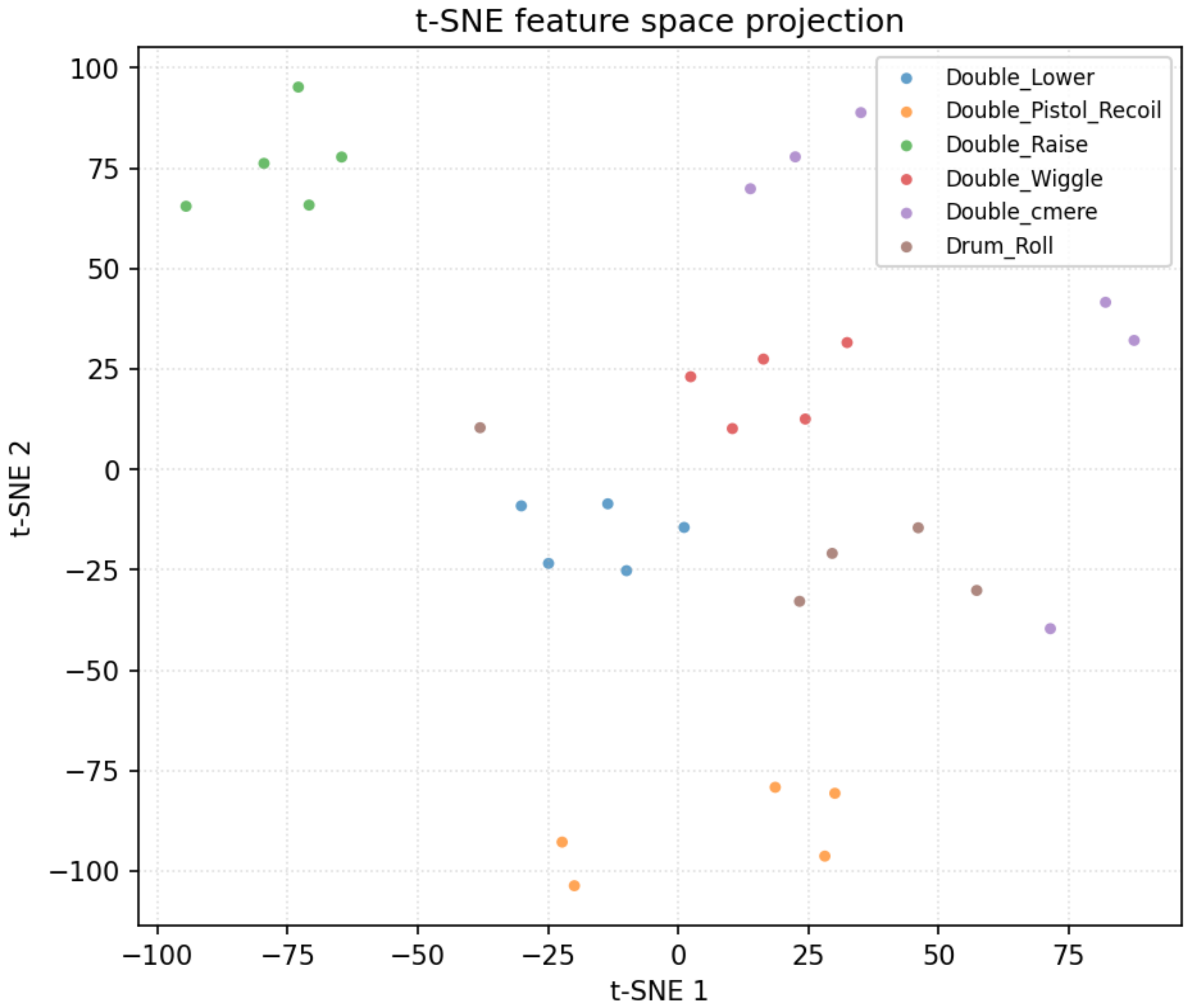

t-SNE feature space projection
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
10
0
−10
−20
−30
−30
−20
−10
0
10
20
30
40
t-SNE 1
t-SNE 2

Data diagnostics — user 3_Anghad

31 trials, 6 classes (filtered + resampled, before per-trial normalisation).

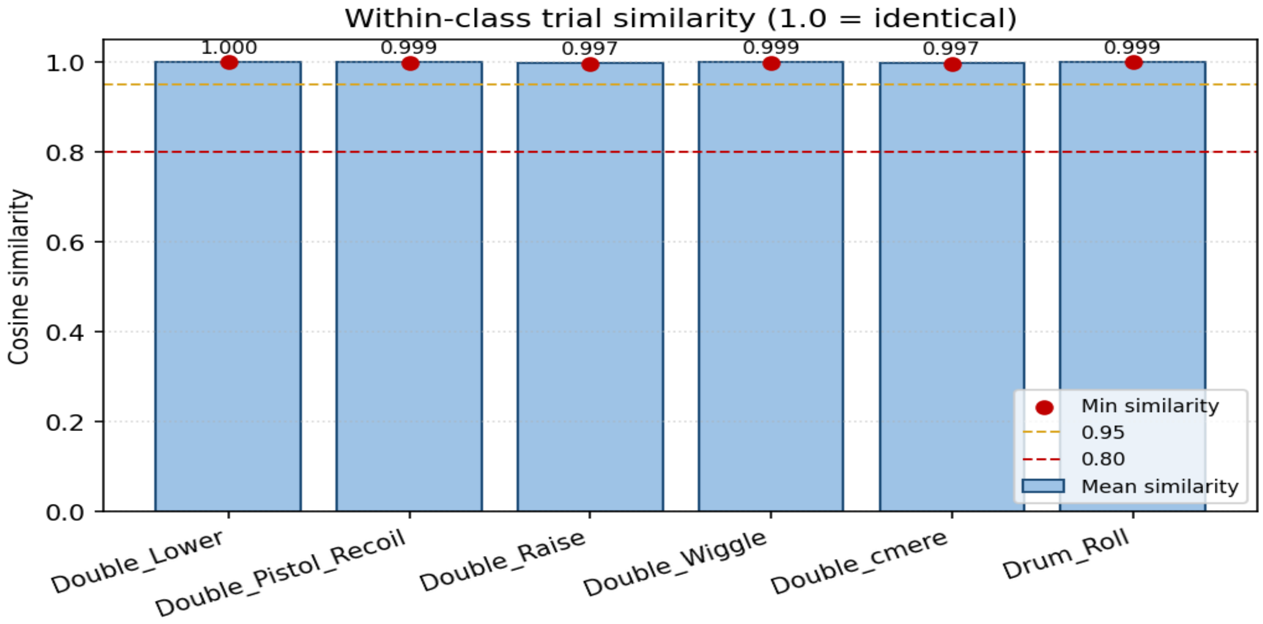


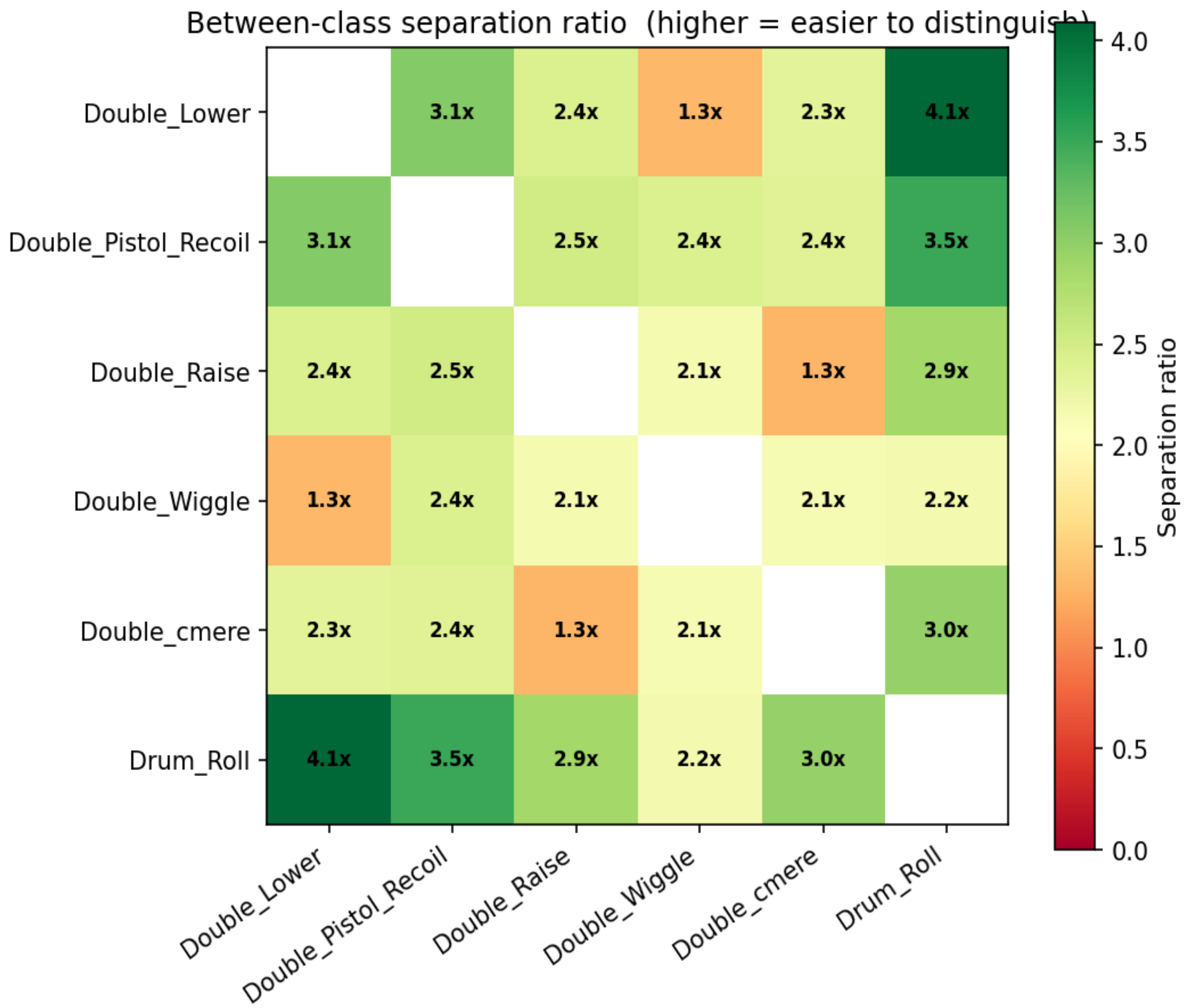


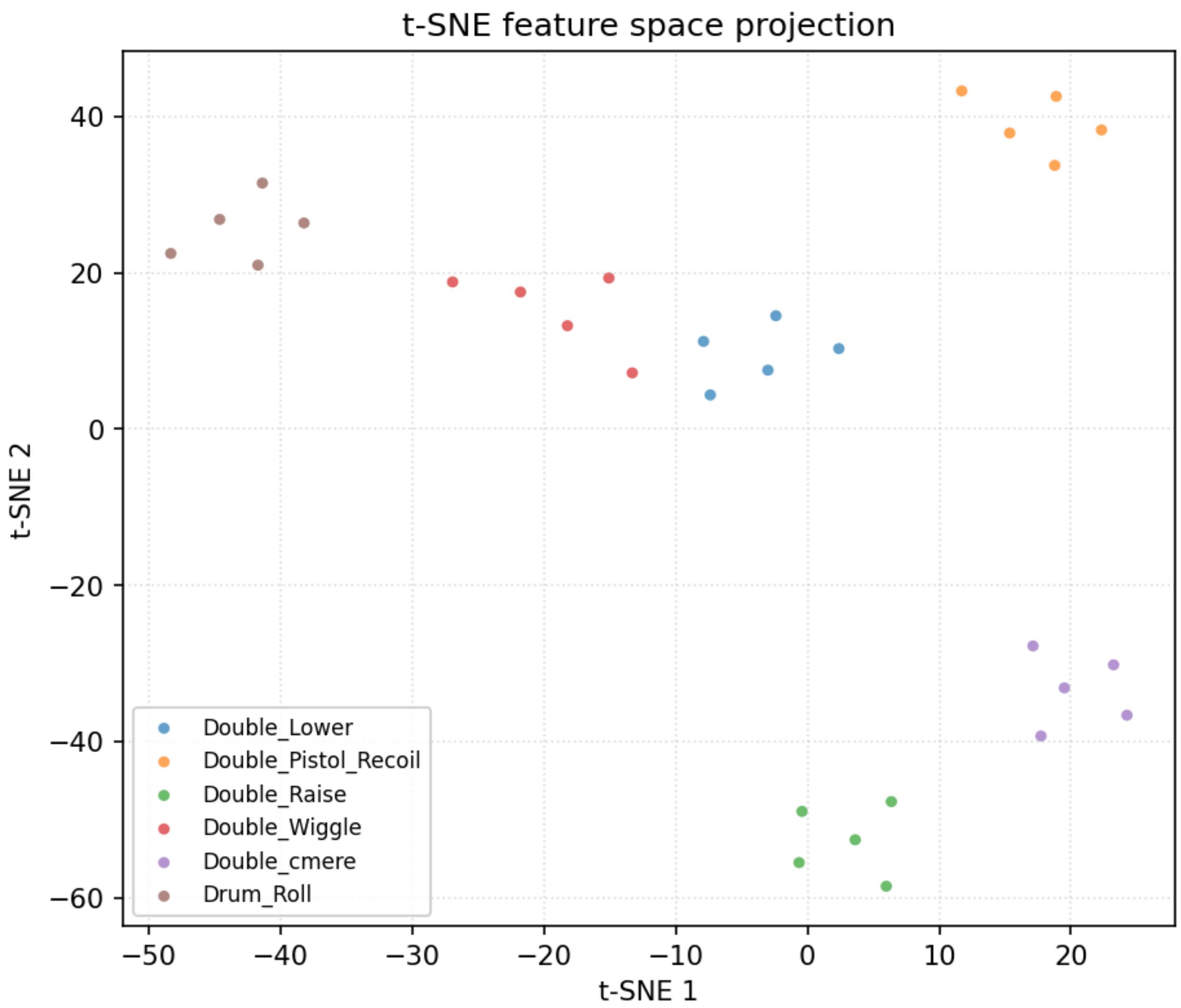

t-SNE feature space projection
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
t-SNE 2
t-SNE 1
100
75
50
25
0
−25
−50
−75
−100
−100
−75
−50
−25
0
25
50
75

Data diagnostics — user 4_Daniel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

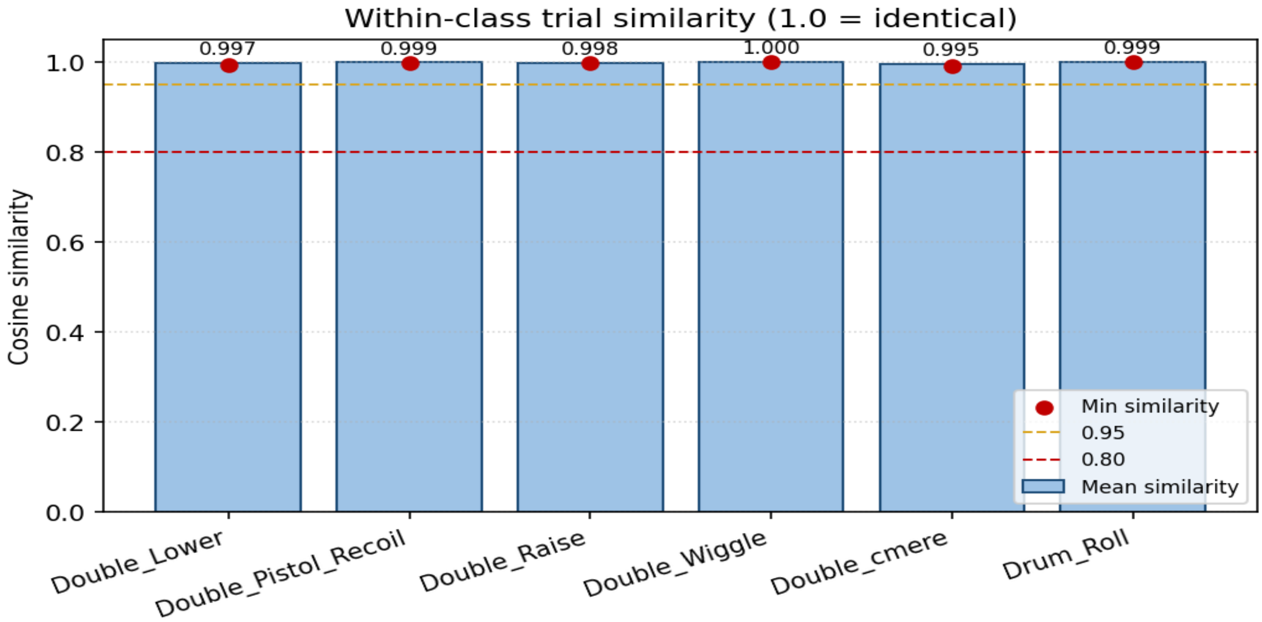


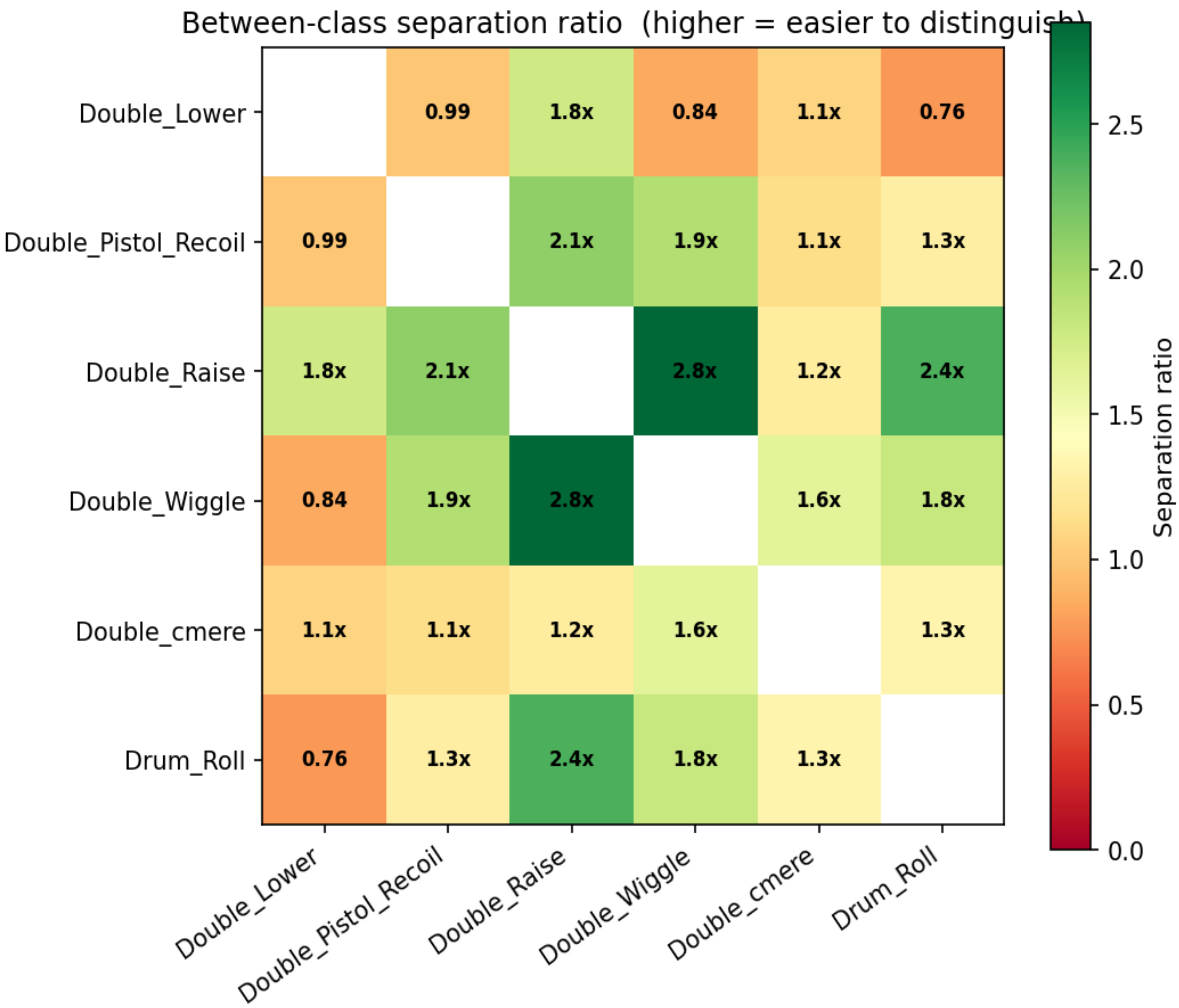



t-SNE feature space projection
t-SNE 2
t-SNE 1
40
20
0
−20
−40
−60
−50
−40
−30
−20
−10
0
10
20
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll

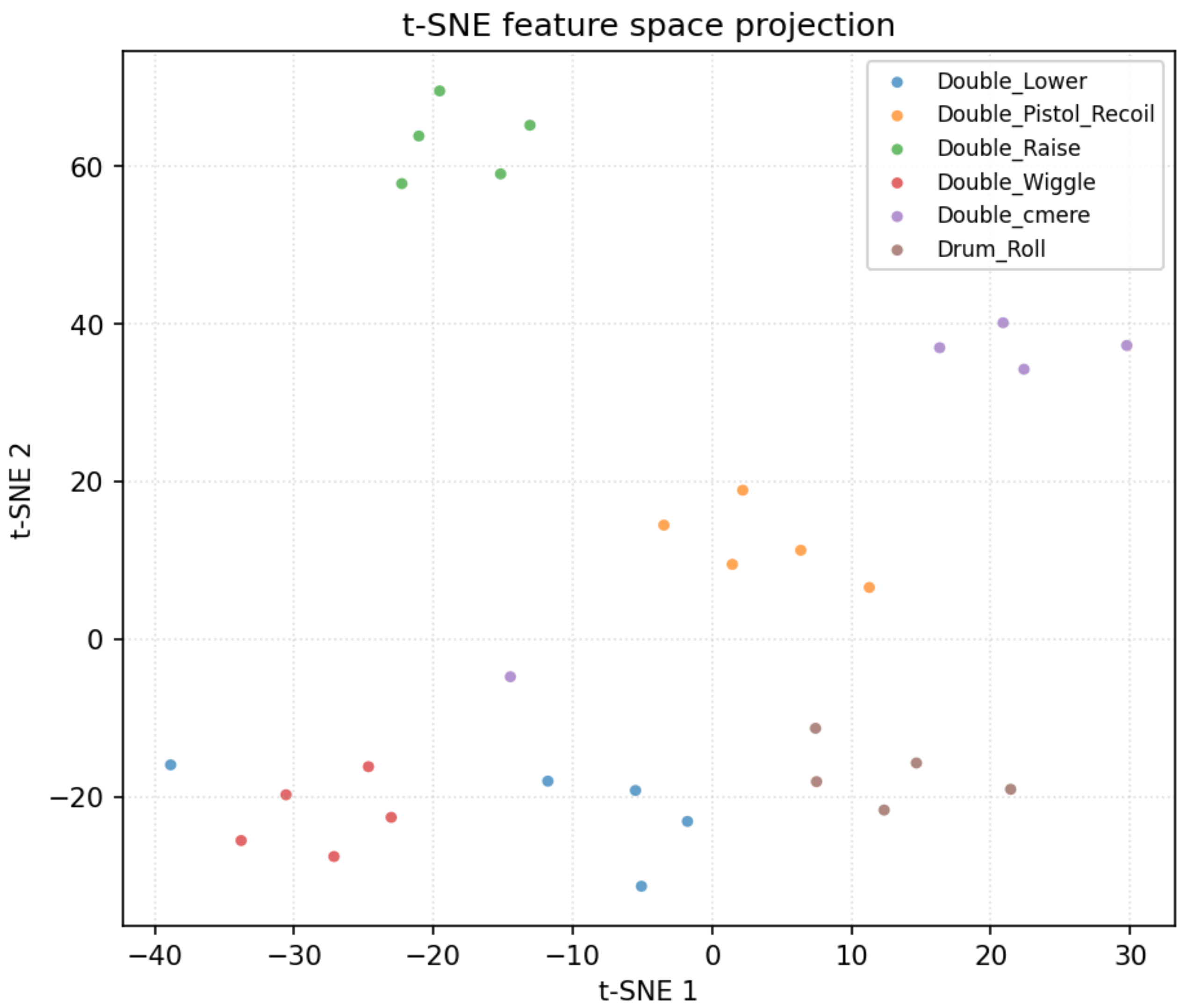
Data diagnostics — user 6_Harry

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).





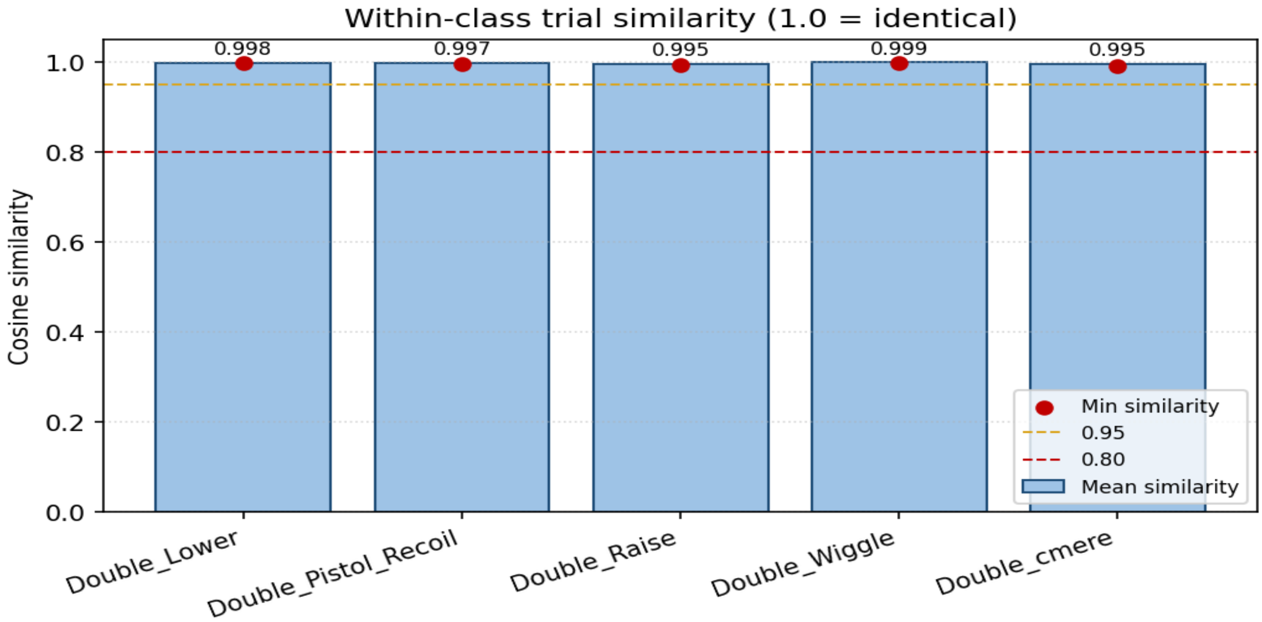
	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll
Double_Lower		0.99	1.8x	0.84	1.1x	0.76
Double_Pistol_Recoil	0.99		2.1x	1.9x	1.1x	1.3x
Double_Raise	1.8x	2.1x		2.8x	1.2x	2.4x
Double_Wiggle	0.84	1.9x	2.8x		1.6x	1.8x
Double_cmere	1.1x	1.1x	1.2x	1.6x		1.3x
Drum_Roll	0.76	1.3x	2.4x	1.8x	1.3x	

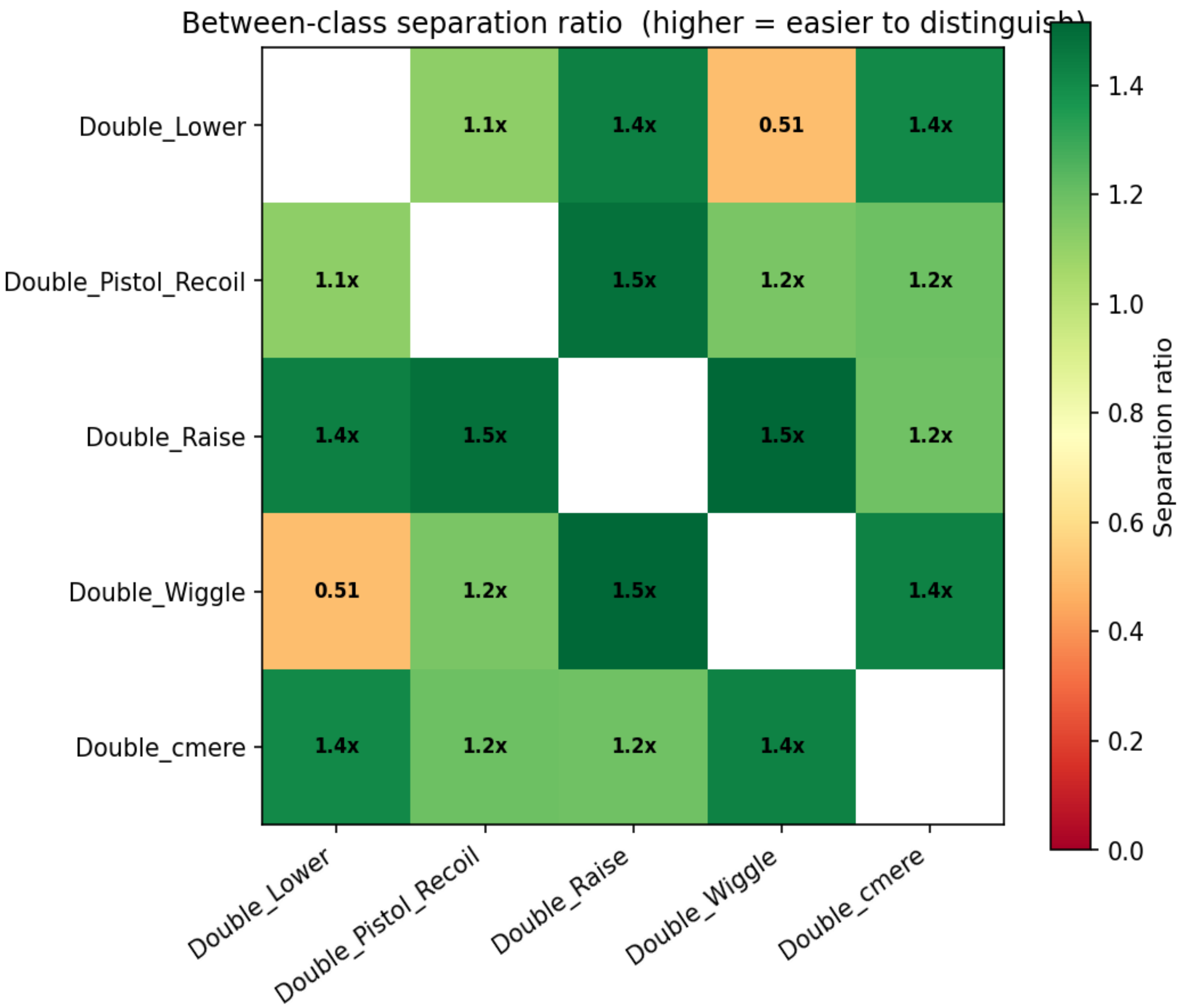


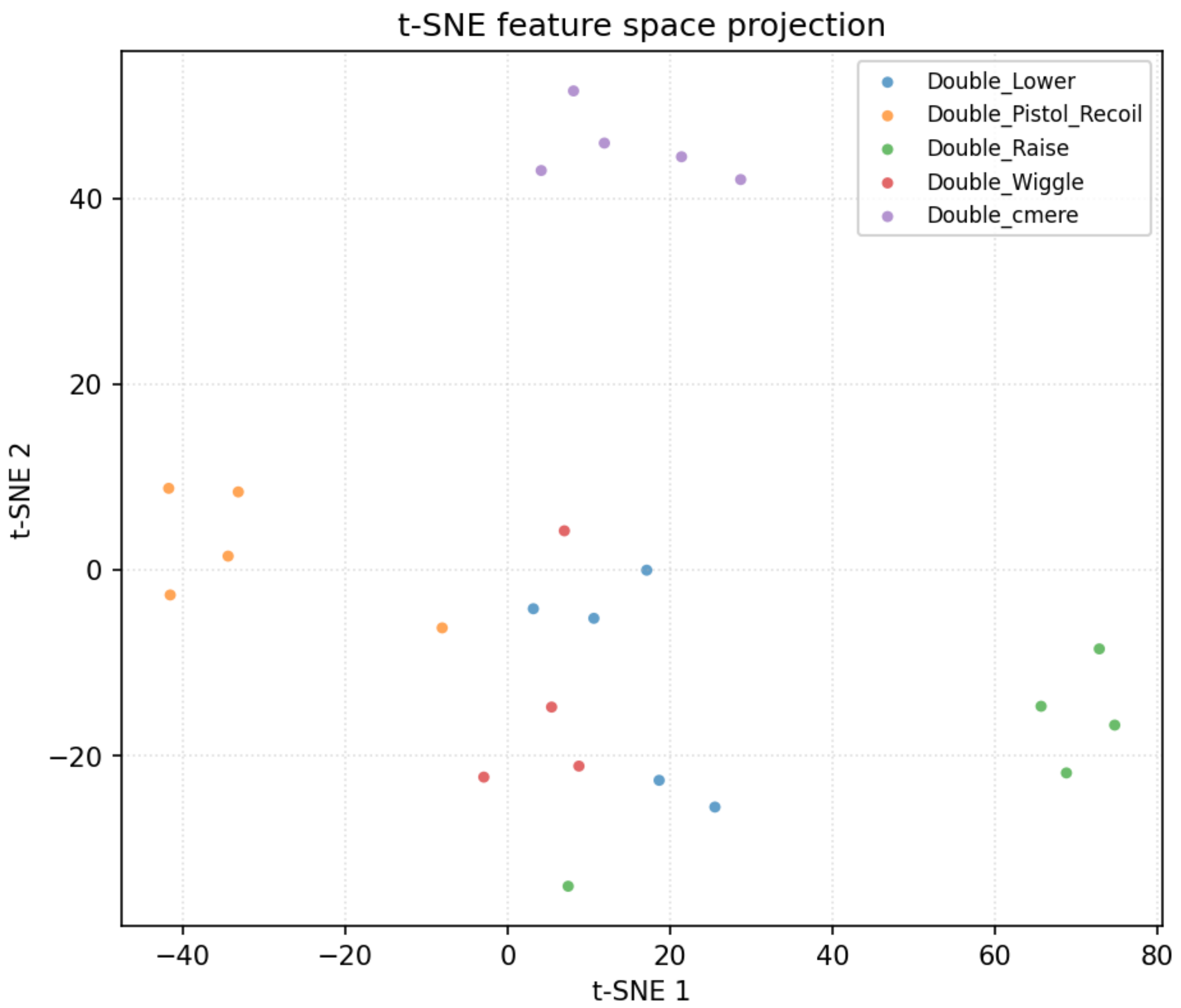
t-SNE feature space projection
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
Drum_Roll
t-SNE 2
t-SNE 1
60
40
20
0
−20
−40
−30
−20
−10
0
10
20
30

Data diagnostics — user 8_Jestin

24 trials, 5 classes (filtered + resampled, before per-trial normalisation).






t-SNE feature space projection
Double_Lower
Double_Pistol_Recoil
Double_Raise
Double_Wiggle
Double_cmere
40
20
0
−20
−40
−20
0
20
40
60
80
t-SNE 1
t-SNE 2

Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	51,750
LOUO mean ± std	0.9718 ± 0.0450
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	325	30	1.0000
2	12_Marcus	14_StephenV2	325	30	0.9333
3	14_StephenV2	15_Tash	325	30	0.9667
4	15_Tash	16_Bella	325	30	1.0000
5	16_Bella	17_Raquel	325	30	1.0000
6	17_Raquel	18_Bridgette	325	30	1.0000
7	18_Bridgette	1_Alan	325	30	1.0000
8	1_Alan	2_Alex	325	30	0.9667
9	2_Alex	3_Anghad	324	30	0.9667
10	3_Anghad	4_Daniel	324	31	1.0000
11	4_Daniel	6_Harry	325	30	1.0000
12	6_Harry	8_Jestin	331	30	0.9667
13	8_Jestin	11_Mansh	331	24	0.8333

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.985	1.000	0.992	65
Double_Pistol_Recoil	0.969	0.954	0.961	65
Double_Raise	0.984	0.954	0.969	65
Double_Wiggle	0.985	1.000	0.992	64
Double_cmere	0.928	0.970	0.948	66
Drum_Roll	1.000	0.967	0.983	60
macro avg	0.975	0.974	0.974	385
weighted avg	0.975	0.974	0.974	385

Confusion matrix (aggregated across folds)